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Multi-access Edge Computing (MEC); Edge Platform Application Enablement

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Multi-access Edge Computing (MEC).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document focuses on the functionalities enabled via the Mp1 reference point between MEC applications and MEC platform, which allows these applications to interact with the MEC system. Service management functionality includes registration/deregistration, discovery and event notifications. Application support functionality includes application start-up, registration and termination, traffic rules, DNS and time of day. It describes the information flows, required information, and specifies the necessary operations, data models and API definitions.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] Void.

[2] ETSI GS MEC 002: "Multi-access Edge Computing (MEC); Use Cases and Requirements".

[3] ETSI GS MEC 003: "Multi-access Edge Computing (MEC); Framework and Reference Architecture".

[4] ETSI GS MEC 010-2: "Multi-access Edge Computing (MEC); MEC Management; Part 2: Application lifecycle, rules and requirements management".

[5] ETSI GS MEC 009: "Multi-access Edge Computing (MEC); General principles, patterns and common aspects of MEC Service APIs".

[6] Void.

[7] Void.

[8] IETF RFC 3986: "Uniform Resource Identifier (URI): Generic Syntax".

[9] IETF RFC 7159: "The JavaScript Object Notation (JSON) Data Interchange Format".

[10] W3C® Recommendation (16 August 2006): "Extensible Markup Language (XML) 1.1 (Second Edition)", edited in place 29 September 2006.

[11] IETF RFC 9110: "HTTP Semantics".

[12] IETF RFC 6455: "The WebSocket Protocol".

[13] IETF RFC 6749: "The Oauth 2.0 Authorization Framework".

[14] Void.

[15] ETSI GS NFV-IFA 007: "Network Functions Virtualisation (NFV) Release 3; Management and Orchestration; Or-Vnfm reference point - Interface and Information Model Specification".

[16] IETF RFC 5789: "PATCH Method for HTTP".

[17] IETF RFC 7386: "JSON Merge Patch".

[18] Void.

[19] Void.

[20] Void.

[21] ETSI TS 129 222: "LTE; 5G; Common API Framework for 3GPP Northbound APIs (3GPP TS 29.222 Release 18)".

[22] Void.

[23] ETSI TS 123 222: "LTE; 5G; Common API Framework for 3GPP Northbound APIs (3GPP TS 23.222 Release 18)".

2.2 Informative references

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The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] IETF RFC 5905: "Network Time Protocol Version 4: Protocol and Algorithms Specification".

[i.2] IEEE 1588™-2019: "IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems".

[i.3] Protocol Buffers Documentation: "Language Guide (proto 3)".

[i.4] OASIS Standard: "MQTT Version 3.1.1", 29 October 2014.

[i.5] gRPC™.

[i.6] OpenAPI™ Specification.

[i.7] IETF RFC 4122: "A Universally Unique Identifier (UUID) URN Namespace".

[i.8] Void.

[i.9] Void.

[i.10] ETSI GR MEC 001: "Multi-access Edge Computing (MEC) Terminology".

[i.11] ETSI TS 129 558: "5G; Enabling Edge Applications; Application Programming Interface (API) specification; Stage 3 (3GPP TS 29.558 Release 17)".

[i.12] ETSI TS 123 558: "5G; Architecture for enabling Edge Applications (3GPP TS 23.558 Release 17)".

[i.13] ETSI TS 129 122: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; T8 reference point for Northbound APIs (3GPP TS 29.122)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI GR MEC 001 [i.10] and the following apply:

API exposing function: entity which provides the service communication entry point for the service APIs

NOTE: As defined in ETSI TS 123 222 [23].

API invoker: entity which invokes the CAPIF or service APIs

NOTE: As defined in ETSI TS 123 222 [23].

Common API framework: framework comprising common API aspects that are required to support service APIs

NOTE: As defined in ETSI TS 123 222 [23].

Northbound API: service API exposed to higher-layer API invokers

NOTE: As defined in ETSI TS 123 222 [23].

Service API: interface through which a component of the system exposes its services to API invokers by abstracting the services from the underlying mechanisms

NOTE: As defined in ETSI TS 123 222 [23].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI GR MEC 001 [i.10] and the following apply:

AEF	API Exposing Function
APF	API Publishing Function
DSCP	Differentiated Services Code Point
NTP	Network Time Protocol
PoP	Point of Presence
PTP	Precision Time Protocol
TC	Traffic Class
ToD	Time of Day

4 Overview

The present document specifies two MEC Platform Application Enablement APIs that support the requirements defined for Multi-access Edge Computing in ETSI GS MEC 002 [2], namely the MEC application support API and the MEC service management API.

Clause 5 introduces the functionalities enabled via the Mp1 reference point between MEC applications and MEC platform. It provides the high level information flows and describes the necessary operations.

The common data types are defined in clause 6, while the corresponding data models and API definitions are specified in clause 7 for the MEC application support API and clause 8 for the MEC service management API. Clause 9 specifies a mapping of the MEC service management API to the 3GPP Common API Framework for northbound APIs (CAPIF) [21].

5 Description of the services (informative)

5.1 Introduction

The MEC platform, as defined in ETSI GS MEC 003 [3], offers an environment where MEC applications may discover, advertise, consume and offer MEC services. Upon receipt of update, activation or deactivation of traffic rules from the MEC platform manager, applications or services, the MEC platform instructs the data plane accordingly. The MEC platform also receives DNS records from the MEC platform manager and uses them to configure a DNS proxy/server.

Via Mp1 reference point between the MEC platform and the MEC applications, as defined in ETSI GS MEC 003 [3], the basic functions are enabled, such as:

- MEC service management:
  - authentication and authorization of producing and consuming MEC services;
  - a means for service producing MEC applications to register/deregister towards the MEC platform the MEC services they provide, and to update the MEC platform about changes of the MEC service availability;
  - a means to notify the changes of the MEC service availability to the relevant MEC application;
  - discovery of available MEC services;
- MEC application support:
  - MEC application start-up procedure;
  - MEC application graceful termination/stop;
  - MEC application registration;
- traffic routing:
  - traffic rules update, activation and deactivation;
- DNS rules:
  - DNS rules activation and deactivation;
- timing:
  - providing access to time of day information;
- transport information:
  - providing information about available transports.

These functions are grouped into those considered to provide MEC application support (i.e. application specific traffic routing, DNS rules and timing, as well as graceful termination/stop) and those that provide MEC service management (i.e. MEC service assistance and associated service transport information).

5.2 Sequence diagrams

5.2.1 General

The following clauses describe how MEC applications and/or MEC services may be supported by the MEC platform via Mp1 reference point. The related sequence diagrams are presented.

5.2.2 MEC application start-up

Figure 5.2.2-1 shows three alternative messages that a MEC application can use to communicate with a MEC platform during the start-up phase of the application instantiation process, steps 5 to 7 in clause 5.3.1 of ETSI GS MEC 010-2 [4].

In this flow, the MEC platform can verify the authenticity of the MEC application with the aid of an AA entity that contains the registration related information about the MEC application in question. For actual authentication, the MEC application uses access token based on Oauth2.0.

MEC platform also has possibility to verify the correctness of the service registration or services query of the MEC application, as it is assumed that MEC platform has received the valid configuration for service consuming and service producing MEC applications. The related information about this MEC application instance (including the required and the optional services, the services to be offered by this application instance and the associated transport dependency, the traffic rules and DNS rules associated with this application instance, etc.) can be compared to those included in the service registration or services query messages, which can be used to determine whether to accept or reject the request.

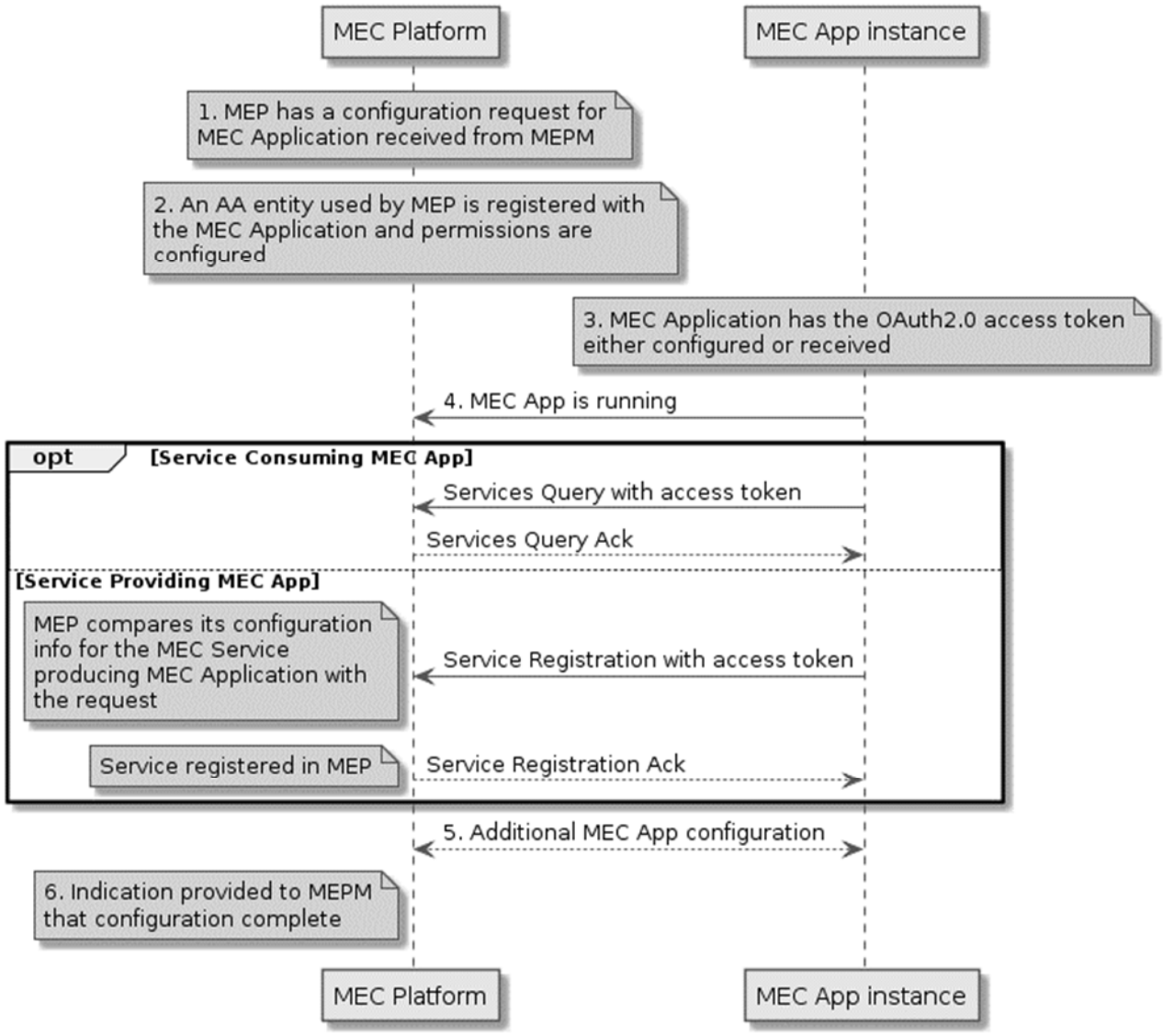


Figure 5.2.2-1: Flow of MEC application start up

MEC application start up procedure, following the MEC application instantiation procedure (as defined in ETSI GS MEC 010-2 [4]), consists of the following steps:

1) MEC platform has received a configuration request from MEC Platform Manager. The configuration request contains detailed information about the parameters related to the MEC application, including the required and the optional services, the services to be offered by this application instance and the associated transport dependency, the traffic rules and DNS rules associated with this application instance, etc.

2) An AA entity associated with the MEC platform has been configured with the MEC application related identity and permissions.

3) MEC application that intends to communicate with MEC platform has the Oauth2.0 access token either received or configured.

4) The MEC application that intends to communicate with MEC sends a "MEC App is running" message towards the MEC platform to confirm that the instantiation and the start-up phase have been successfully completed. If the application receives an error response with 409 status code from the platform, it should retry sending such message for a configurable period of time. This behaviour increases robustness to race-conditions in the instantiation process, in particular when the MEC platform has not yet received the configuration request from MEC Platform Manager, or the MEC platform is still processing the configuration request.

   Depending on the nature of the MEC application and its intended use of MEC services, after the successful response received the MEC application may apply one or both of the following options:

   a) Option 1:

      Send services query to the MEC platform (MEC Application that consumes MEC Services). The services query request contains the access token.

   b) Option 2:

      Send a service registration request to the MEC platform (MEC application that provides MEC service(s)). The service registration request contains the access token. The MEC platform then compares the configuration it has for the service producing MEC application with the request, and if acceptable, registers the MEC service and returns a service registration acknowledgement.

NOTE 1: It is out of scope how a MEC application instance discovers a MEC platform. In practice, this may be statically configured or dynamically discovered via e.g. DNS.

5) If applicable, additional configuration on the MEC services may be performed between the MEC platform and MEC application.

   The MEC system may also pre-configure (not through Mp1) the MEC application instance with necessary parameters, for example:

   - the information needed to access the required services;
   - the availability of the optional services;
   - the information needed to access the available optional services.

   The additional procedures via Mp1 that are related to this step include, when required, "Traffic rule activation/deactivation/update" as defined in clause 5.2.7, and "DNS rule activation/deactivation" as defined in clause 5.2.8. And the MEC application instance may update the MEC platform with the information about the available produced MEC services as defined in clause 5.2.4.

6) MEC platform sends an indication to MEC Platform Manager once the configuration is complete. This message is not further specified in the present document.

NOTE 2: The options 4a) and 4b) present different messages that can be sent by a MEC application. As MEC application can both consume and provide MEC service(s), it is possible that such MEC application performs both services query and service registration steps, in any order.

5.2.3 MEC application graceful termination/stop

Figure 5.2.3-1 shows a flow for MEC application instance graceful termination/stop (as defined in ETSI GS MEC 010-2 [4]). After the MEC platform receives a request to terminate or stop a MEC application instance the MEC platform notifies the MEC application instance that it will be terminated or stopped soon if graceful termination/stop is required. In the notification, the MEC platform indicates to the MEC application instance the time interval for the application to perform application-specific termination/stop actions. The time interval is set according to the graceful termination/stop timeout value in the received request to terminate or stop. When this timer expires, the MEC platform continues the termination flow of the MEC application instance or stop MEC application instance flow by, e.g. deactivating the traffic rules and DNS rules, removing the MEC application instance from the list of instances to be notified about service availability, removing the services provided by the MEC application instance from the service registry, sending service availability notification to the MEC applications that consumes the services produced by the terminating/stopping MEC application instance, etc.

The MEC application instance has the option to, before the timer expires, inform the MEC platform that it is ready to be terminated/stopped after it has finished any application level related actions. Upon receipt of this information, the MEC platform continues the flow to terminate or stop the MEC application instance. The service producing MEC application instance should also deregister its produced MEC service(s) towards the MEC platform before the timer expires. Upon receipt of the request, the MEC platform deregisters the MEC service(s).

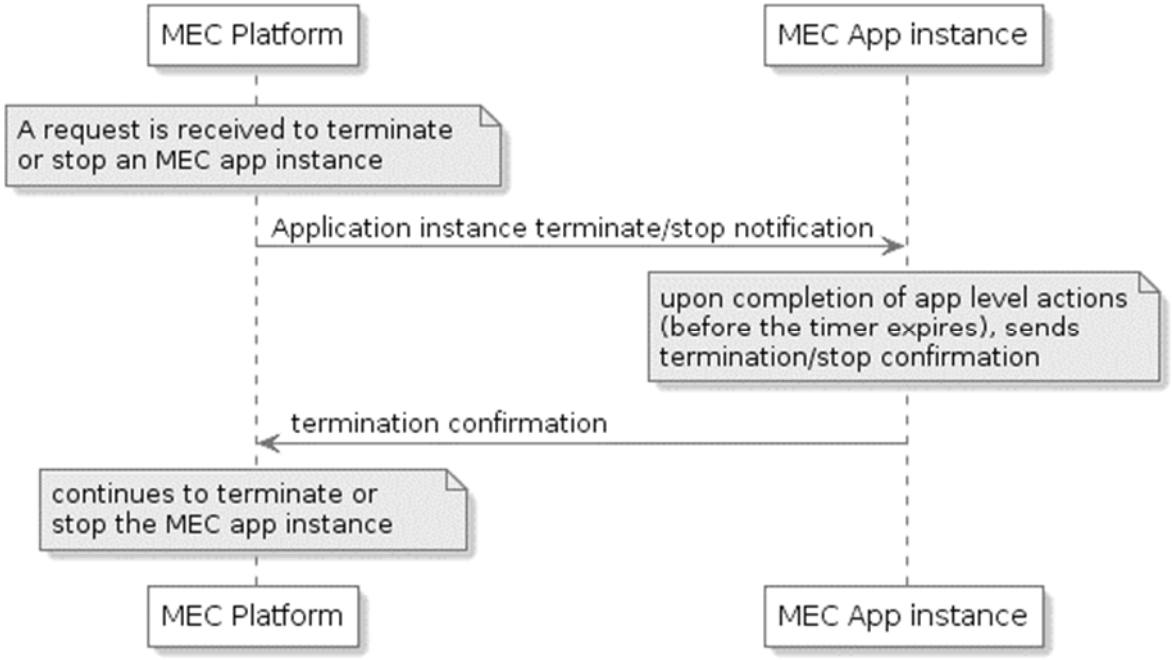


Figure 5.2.3-1: Example flow of MEC application instance graceful termination/stop

5.2.4 Service availability update and new service registration

When a MEC service is registered by the service producing MEC application, the authorized relevant applications (e.g. the applications that indicate the service as "optional" or "required") will be notified about the newly available service. Moreover, the authorized relevant applications will also be notified about the service availability changes of that service.

Figure 5.2.4-1 shows two cases. In the 1st case a MEC application instance informs the MEC platform that the service(s) provided by this application instance become available for the first time (service registration); and then the MEC platform notifies the authorized relevant application instances (e.g. the applications that indicate the service(s) as "optional" or "required") about the newly available service(s). As part of service registration, the relevant information about the service is provided to the platform, and the service is bound to a transport that is either provided by the MEC platform, or by the application itself.

In the 2nd case the service producing MEC application instance updates the MEC platform about the status change of the produced MEC services; and the MEC platform notify the authorized relevant application instances about the service availability changes.

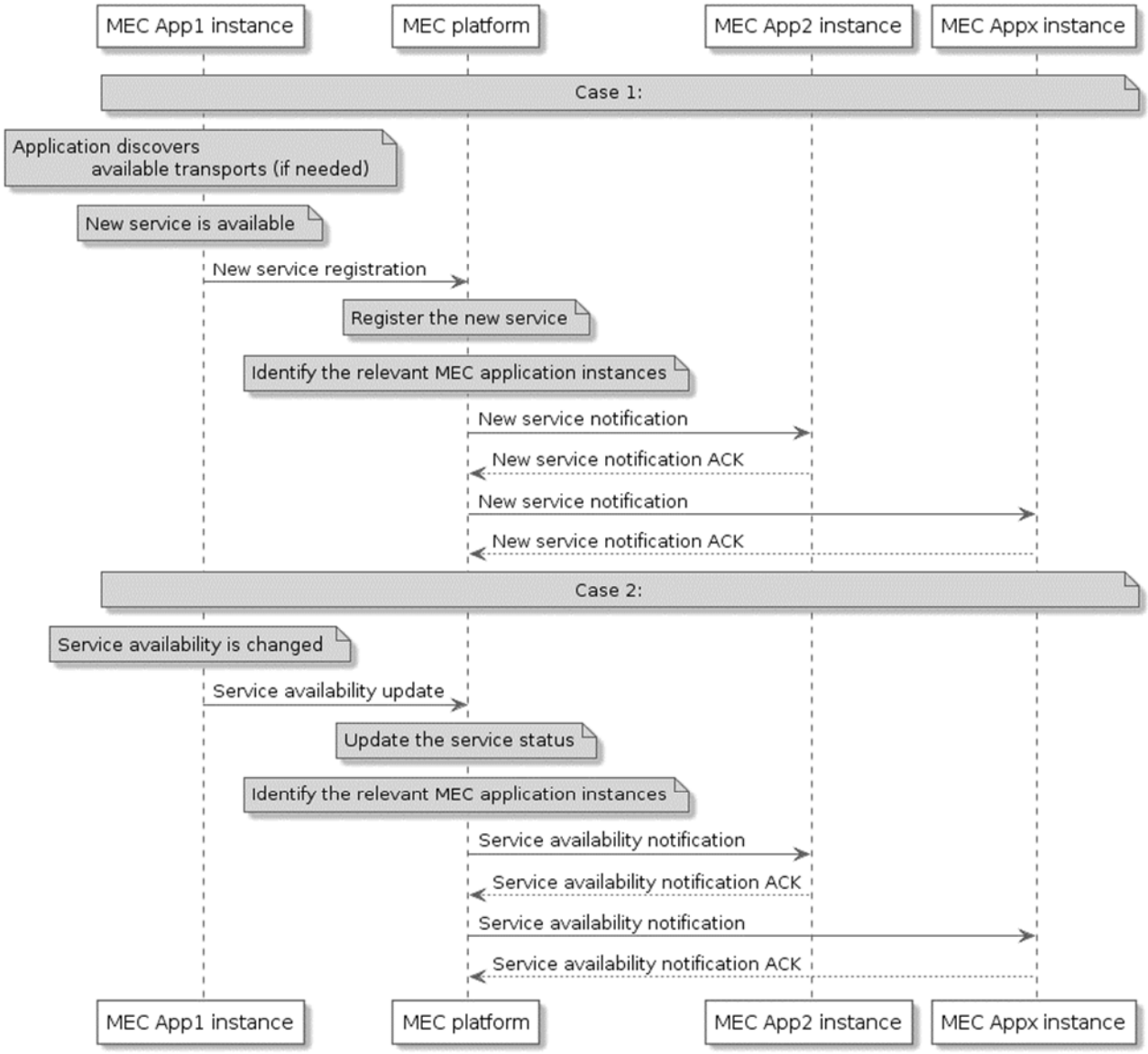


Figure 5.2.4-1: Flow of new service registration and service availability update

In the 1st case the new service registration procedure consists of the following steps:

1) If the application intends to use a transport that is provided by the MEC platform, it discovers the available transports first, and selects one (or more) for use with the new service.

2) After a new MEC service becomes available, the service producing MEC application instance sends new service registration message to the MEC platform.

3) MEC platform registers the new service with the service registry. This step is not to be further specified.

4) MEC platform then identifies the relevant MEC application instance for this update (e.g. the applications that indicate the service as "optional" or "required"), and sends new service notifications to the relevant application instances. When supported and the service instance can be consumed by MEC applications running on other MEC hosts, the MEC platform identifies the relevant MEC platforms for this update, and informs them about the changes in service availability by means that may be outside the scope of the present document. The relevant MEC platforms then flag the MEC service instance as running on the other MEC host and send new service notifications to the relevant application instances.

5) The MEC application instances, optionally, acknowledge the notification.

In the 2nd case MEC service availability update procedure consists of the following steps:

1) When a MEC service changes its availability, the service producing MEC application instance sends service availability update message to the MEC platform.

2) MEC platform updates the service registry. This step is not to be further specified.

3) MEC platform then identifies the relevant MEC application instance for this update (e.g. the applications that indicate the service as "optional" or "required"), and sends service availability notifications to the relevant application instances. If supported and the service can be consumed by MEC applications running on other MEC hosts, the MEC platform identifies the relevant MEC platforms for this update, and informs them about the changes in service availability by means that may be outside the scope of the present document. The relevant MEC platforms then send service availability notifications to the relevant application instances.

4) The MEC application instances, optionally, acknowledge the notification.

NOTE 1: In the present document it is not specified on how the MEC platform determines the relevant remote MEC platforms in steps 4 (1st case) and step 3 (2nd case).

NOTE 2: In the present document it is not specified on how MEC orchestrator is kept informed of the service status updates in remote MEC platforms.

5.2.5 Service availability query

Figure 5.2.5-1 shows a scenario where a MEC application instance sends a request to receive information on the availability of a MEC service or a list of MEC services. Typically a MEC application may only query about the MEC service(s) that it has indicated as "optional" or "required".

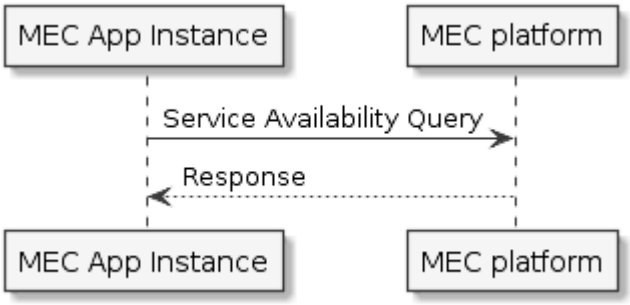


Figure 5.2.5-1: Flow of MEC application requesting service availability information

MEC application requesting service availability information, as illustrated in figure 5.2.5-1, consists of the following steps:

1) MEC application instance sends a request to query the availability of a MEC service or a list of MEC services. Typically a MEC application instance may only query about the MEC service(s) that it has indicated as "optional" or "required".

2) MEC platform responds with the message body containing the information about the available service(s), including the information needed to access the available service(s). Note that the service availability information is updated by the service producing MEC application instances to the MEC platform.

5.2.6 Managing subscription to event notifications

5.2.6.1 Introduction

A subscription is required for event notifications that are sent from the MEC platform.

A service availability notification is sent in the following two cases as described in clause 5.2.4:

- When a MEC service is made available by the service producing MEC application, the authorized relevant applications (e.g. the applications that indicate the services as "optional" or "required") will be notified about the newly available service.
- The authorized relevant applications will also be notified about the service availability changes.

An application instance terminate/stop notification is sent in the following two cases as described in clause 5.2.3:

- The MEC platform has received a request for graceful termination of a MEC application instance.
- The MEC platform has received a request for graceful stop of a MEC application instance.

This clause describes the sequence diagram of two related procedures:

- Subscribing to event notifications.
- Unsubscribing from event notifications.

5.2.6.2 Subscribing to event notifications

Figure 5.2.6.2-1 shows the message flow for subscribing to event notifications.

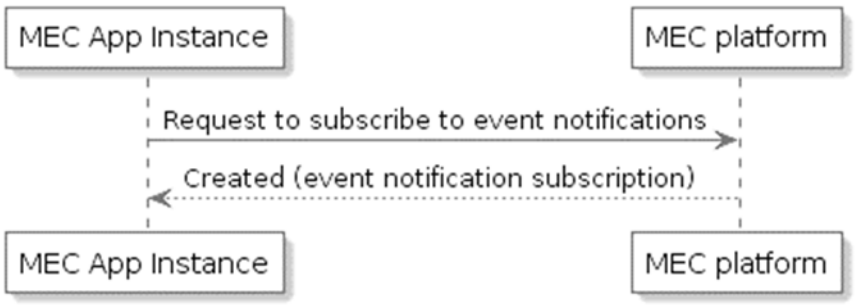


Figure 5.2.6.2-1: Flow of Subscribing to event notifications

MEC application requesting event notifications subscription consists of the following steps:

1) MEC application instance sends a request to subscribe to event notifications. In case of service availability event notifications, typically a MEC application instance may only subscribe to availability event notifications of the MEC service(s) that it has indicated as "optional" or "required".
2) MEC platform responds with the message body containing the created subscription to the event notifications.

5.2.6.3 Unsubscribing from event notifications

Figure 5.2.6.3-1 shows the message flow for unsubscribing from event notifications.

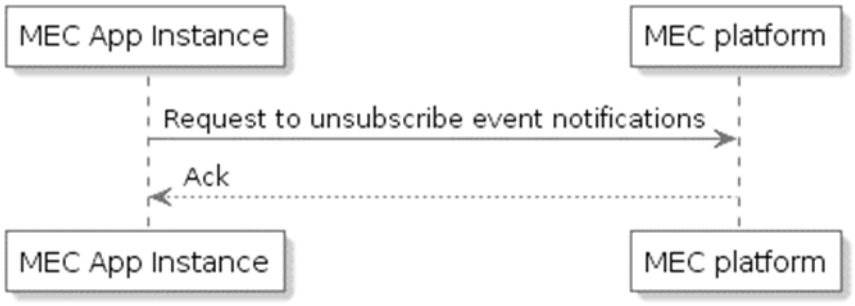


Figure 5.2.6.3-1: Flow of unsubscribing from event notifications

MEC application requesting to unsubscribe from the event notifications consists of the following steps:

1) MEC application instance sends a request to unsubscribe from the event notifications.

2) MEC platform responds with an acknowledgement.

5.2.6a Receiving event notifications on MEC service availability

Figure 5.2.6a-1 shows the scenario where the MEC Platform sends event notification on MEC service availability to the MEC App instance.

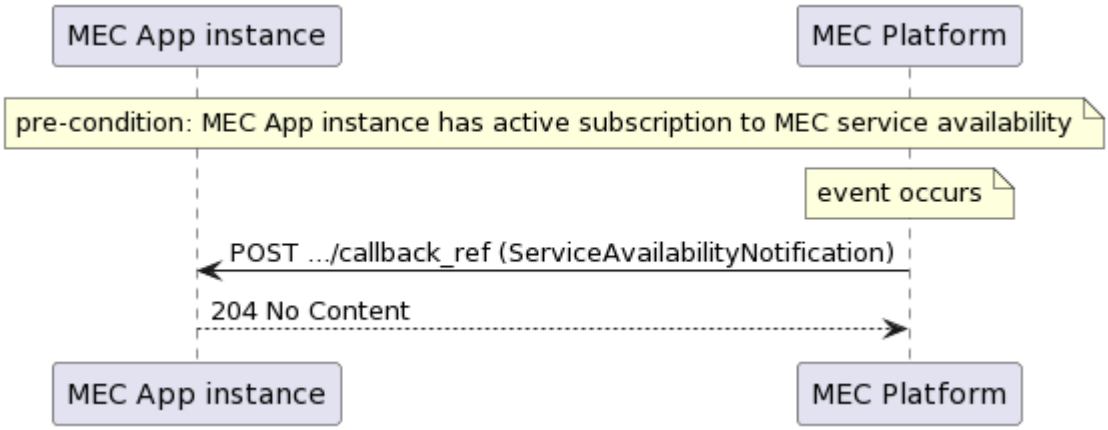


Figure 5.2.6a-1: Flow of receiving event notifications on MEC service availability

Receiving event notifications on MEC service availability, as illustrated in figure 5.2.6a-1, consists of the following steps:

1) MEC Platform sends a POST, with the message content containing ServiceAvailabilityNotification data structure, to the callback reference address included by the MEC App instance in the MEC service availability event subscription.

2) MEC App instance sends a "204 No Content" response to MEC Platform.

5.2.6b Receiving event notifications on MEC application instance termination

Figure 5.2.6b-1 shows the scenario where the MEC Platform sends event notification on MEC application instance termination to the MEC App instance.

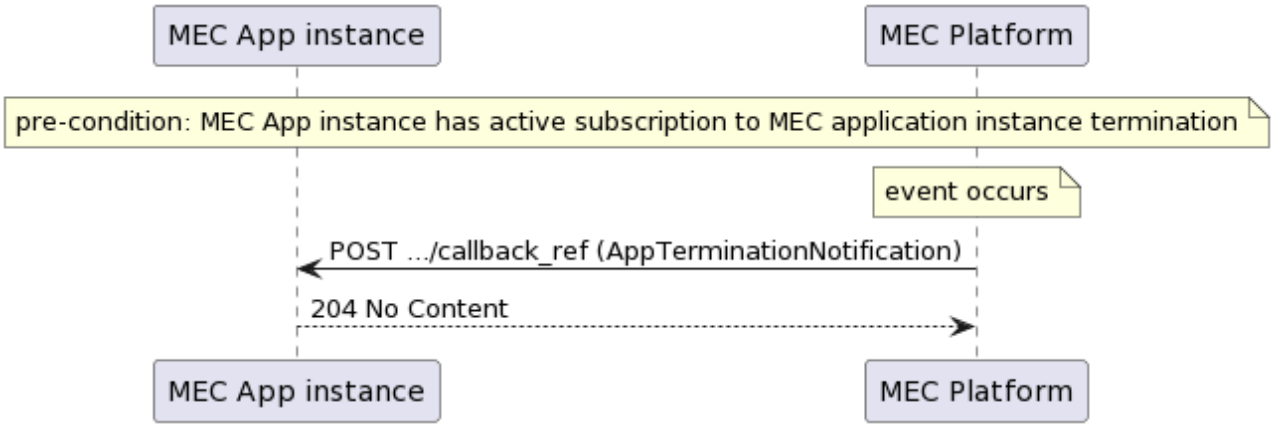


Figure 5.2.6b-1: Flow of receiving event notifications on MEC application instance termination

Receiving event notifications on MEC application instance termination, as illustrated in figure 5.2.6b-1, consists of the following steps:

1) MEC Platform sends a POST, with the message content containing AppTerminationNotification data structure, to the callback reference address included by the MEC App instance in the MEC application instance termination event subscription.

2) MEC App instance sends a "204 No Content" response to MEC Platform.

5.2.7 Traffic rule activation/deactivation/update

Figure 5.2.7-1 shows a flow for traffic rule activation, deactivation, and update. The MEC application instance may request the MEC platform to activate or deactivate a traffic rule(s). The MEC application instance may request the MEC platform to update the parameters of an existing traffic rule(s).



Figure 5.2.7-1: Flow of traffic rule activation/deactivation/update

Traffic rule activation/deactivation/update flow consists of the following steps:

1) The MEC application instance sends traffic rule activation/deactivation/update request to the MEC platform. The message identifies one or multiple traffic rules that will be activated, deactivated or updated. If the request is authorized, the MEC platform may update the data plane via Mp2 reference point, which is out of the scope of the present document.

2) The MEC platform sends response to the MEC application instance to indicate the results of the operation.

5.2.8 DNS rule activation/deactivation

Figure 5.2.8-1 shows a DNS rule activation/deactivation flow. The MEC application instance may request the MEC platform to activate or deactivate a DNS rule(s). If the request is authorized and the MEC platform succeeds in finding, based on the information contained in the request, the corresponding DNS rule(s) that have been pre-configured and authenticated by the MEC management, the MEC platform will install or remove the DNS rule(s) into or from the DNS server/proxy.

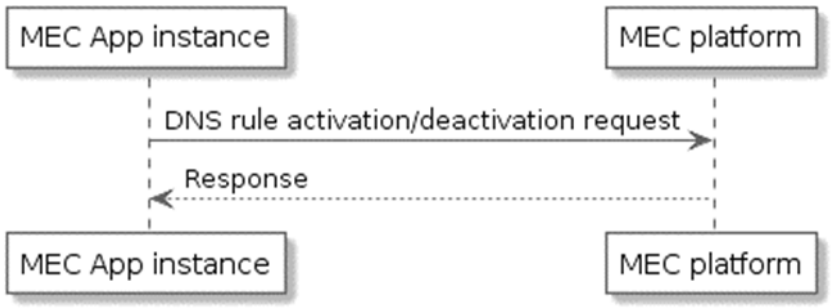


Figure 5.2.8-1: Flow of DNS rule activation/deactivation

DNS activation/deactivation flow consists of the following steps:

1) The MEC application instance sends DNS activation/deactivation request to the MEC platform. The request includes the DNS rule(s) to be activated or deactivated. If the request is authorized and the MEC platform succeeds in finding, based on the information contained in the request, the corresponding DNS rule(s) that have been pre-configured and authenticated by the MEC management, the platform will install or remove the DNS rule(s) from the DNS server/proxy.

2) The MEC platform sends response to the MEC application instance. The response contains the result (success or failure) of the DNS rule activation/deactivation.

5.2.9 Transport information query

Providing a MEC service implies the use of a transport to deliver it to the MEC applications that consume it. Examples of transports are REST-HTTP, and message passing systems that support the Publish-Subscribe mode for the communication between MEC application instances and the MEC platform, or between MEC application instances. Any transport other than REST-HTTP is not further specified in the present document. However, transport information query provides a standardized means to the applications to discover the available transports. Figure 5.2.9-1 shows a scenario where the MEC application instance sends a request to receive information on available transports.

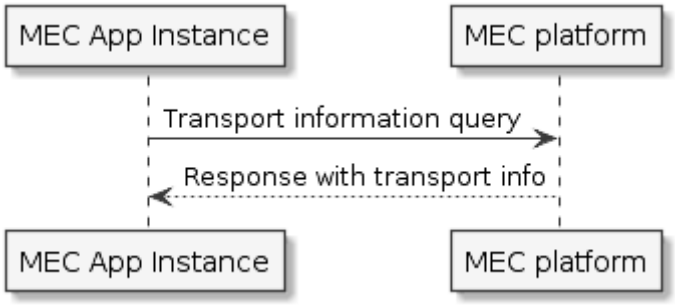


Figure 5.2.9-1: Flow of MEC application requesting transport information

MEC application instance requesting transport information, as illustrated in figure 5.2.9-1, consists of the following steps:

1) MEC application instance sends a request to query the information about transports provided by the platform.

2) MEC platform responds with the message body containing the transports information.

5.2.10 Time of Day (ToD)

5.2.10.1 Introduction

MEC applications may require TOD information for notifications, logs and special events time notions, packets receipt and transmit timestamping and other needs depending on application purpose.

Required TOD accuracy strongly depends on the application itself. Low accuracy TOD information may be provided to the application by use of simple procedure of current time retrieval from the platform. Higher TOD accuracy may be achieved by use of special protocols that allows timing transfer over packet networks, such as NTP specified in IETF RFC 5905 [i.1] or PTP specified in IEEE 1588™ [i.2]. In case of use of packet timing protocols it is assumed that a MEC application will run NTP client or PTP client (referred to as "slave" in [i.2]) while the NTP server(s) or PTP server(s) (referred to as "masters" in [i.2]) may run either by the MEC platform itself or by other facilities for which the application has network connectivity.

This clause specifies two TOD related information exchange flows:

- "Get platform time" flow to get MEC platform current time of day.
- Optional "Timing capabilities query" flow to retrieve information regarding available packet timing facilities.

5.2.10.2 Get platform time

Figure 5.2.10.2-1 shows the flow for getting platform time.

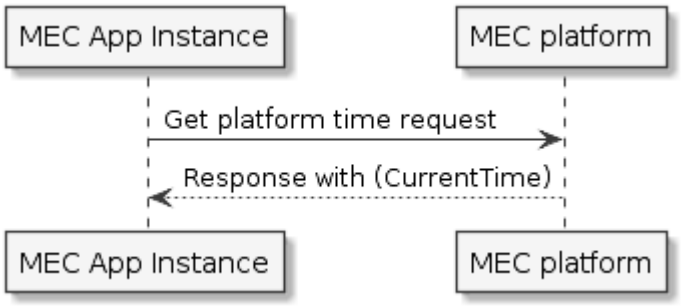


Figure 5.2.10.2-1: Flow of MEC application requesting platform time

Get platform time flow consists of the following steps:

1) The MEC application instance sends the get platform time request to the MEC platform.

2) MEC platform responds with the message body containing CurrentTime.

5.2.10.3 Timing capabilities query flow

Figure 5.2.10.3-1 shows a flow for timing capabilities query.

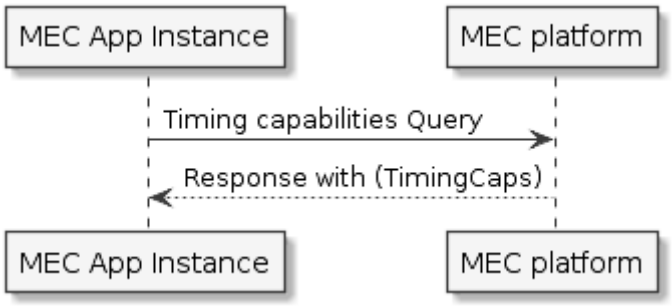


Figure 5.2.10.3-1: Flow of timing capabilities query

Timing capabilities query flow consists of the following steps:

1) The MEC application instance sends the timing capabilities query request to the MEC platform.

2) MEC platform responds with the message body containing TimingCaps.

5.2.11 Service deregistration

Figure 5.2.11-1 shows a scenario where a MEC application instance that provides MEC service(s) sends a service deregistration request to the MEC platform.

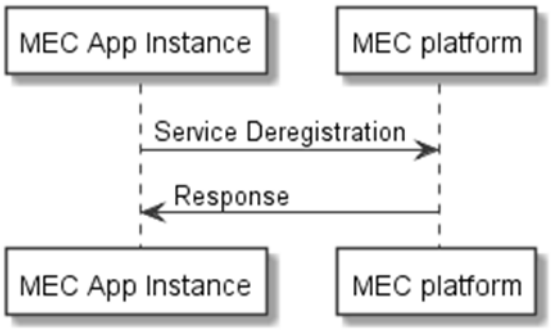


Figure 5.2.11-1: Flow of MEC service deregistration

MEC service registration, as illustrated in figure 5.2.11-1, consists of the following steps:

1) MEC application instance sends a request to the MEC platform to deregister the MEC service it provides.

2) The MEC platform deregisters the MEC service and returns a service deregistration acknowledgement.

5.2.12 Service heartbeat

Figure 5.2.12-1 shows a scenario where a MEC service instance sends a heartbeat message to the MEC platform.

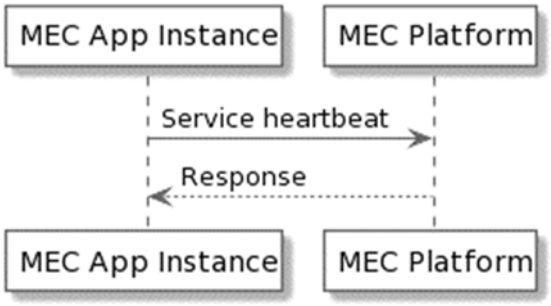


Figure 5.2.12-1: Flow of MEC service deregistration

Each MEC service instance that has previously registered in MEC platform and is configured for heartbeat sends heartbeat message to the MEC platform periodically in order to show that the MEC service instance is still operational.

The time interval at which the MEC platform is contacted is deployment-specific, and is indicated by the MEC platform to the MEC service instance in a successful service registration.

When the MEC platform has not received the heartbeat for a configurable amount of time, the MEC platform considers that the service instance can no longer be discovered. The MEC platform notifies the MEC service consumers subscribed to receive notifications of status change of the MEC service instance.

MEC service heartbeat, as illustrated in figure 5.2.12-1, consists of the following steps:

1) MEC service instance sends a heartbeat message to the MEC platform periodically.

2) The MEC platform returns a service heartbeat acknowledgement.

5.2.13 MEC application registration

5.2.13.1 Introduction

This set of procedures is optional, i.e. it is up to application developer to decide if registration is necessary.

The application registration procedure allows an authorized MEC application instance to provide its information to the MEC platform.

NOTE 1: The application needs to be instantiated before it can start registration procedure.

NOTE 2: For the application instance that is not instantiated by the MEC Management, the registration can ensure the application instance is discoverable. If there is a change in the requirements or to the information of an MEC application instance, the authorized MEC application instance uses the application registration update procedure to update the MEC platform.

The authorized MEC application instance uses the application de-registration procedure to remove its information from the MEC platform.

This clause specifies three MEC application registration related information flows:

- "Application registration" flow to register an MEC application instance to a MEC platform.
- "Application registration update" flow to update an existing MEC application instance registration to a MEC platform.
- "Application de-registration" flow to cancel an existing MEC application registration to a MEC platform.

5.2.13.2 Application registration

Figure 5.2.13.2-1 shows a scenario where a MEC application instance sends an application registration request to the MEC platform.

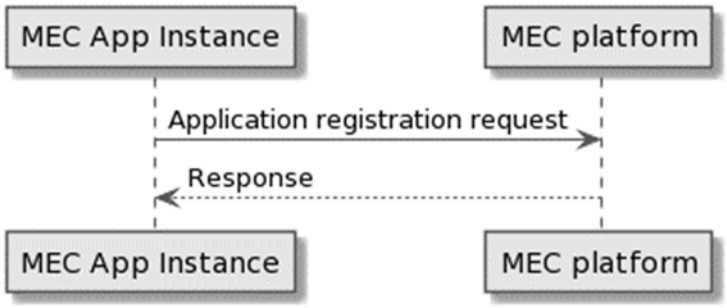


Figure 5.2.13.2-1: Flow of MEC application registration

MEC application registration, as illustrated in figure 5.2.13.2-1, consists of the following steps:

1) MEC application instance sends a request to the MEC platform to register itself.
2) The MEC platform registers the MEC application instance and returns an application registration acknowledgement.

5.2.13.3 Application registration update

Figure 5.2.13.3-1 shows a scenario where a MEC application instance sends an application registration update request to the MEC platform.

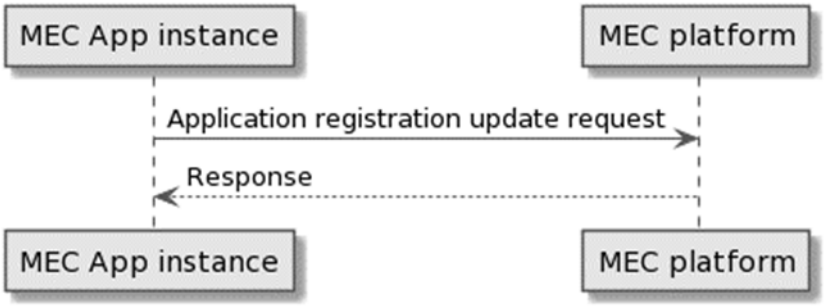


Figure 5.2.13.3-1: Flow of MEC application registration update

MEC application registration update, as illustrated in figure 5.2.13.3-1, consists of the following steps:

1) MEC application instance sends a request to the MEC platform to update an existing application registration.

2) The MEC platform updates the MEC application instance registration and returns an application registration update acknowledgement.

5.2.13.4 Application deregistration

Figure 5.2.13.4-1 shows a scenario where a MEC application instance sends an application deregistration request to the MEC platform.

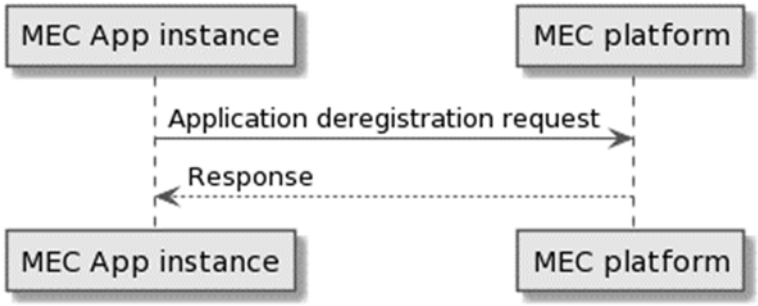


Figure 5.2.13.4-1: Flow of MEC application deregistration

MEC application deregistration, as illustrated in figure 5.2.13.4-1, consists of the following steps:

1) MEC application instance sends a request to the MEC platform to remove an existing application registration.

2) The MEC platform removes the information on the MEC application instance registration and returns an application deregistration acknowledgement.

6 Common data types

6.1 Introduction

The following clauses define the data types common to the APIs specified in the present document.

6.2 Resource data types

6.2.1 Introduction

This clause defines data structures to be used in resource representations.

6.2.2 Type: SubscriptionLinkList

This type represents a list of links related to currently existing subscriptions for a MEC application instance. This information is returned when sending a request to receive current subscriptions.

Table 6.2.2-1: Attributes of the SubscriptionLinkList

Attribute name	Data type	Cardinality	Description
_links	Structure (inlined)	1	Object containing hyperlinks related to the resource.
>self	LinkType	1	Self-referring URI.
>subscriptions	array(Structure (inlined))	0..N	The MEC application instance's subscriptions.
>>href	Uri	1	URI referring to the subscription.
>>subscriptionType	String	1	Type of the subscription. The values are as defined in the "subscriptionType" attribute for each different Mp1 event subscription data type.

6.3 Referenced structured data types

6.3.1 Introduction

This clause defines data structures that are referenced from multiple APIs specified in the present document.

6.3.2 Type: LinkType

This type represents a type of link and may be referenced from data structures.

Table 6.3.2-1: Attributes of the LinkType

Attribute name	Data type	Cardinality	Description
href	Uri	1	URI referring to a resource.

7 MEC application support API

7.1 Data model

7.1.1 Introduction

Clauses 7.1.2 to 7.1.6 specify the data types that are used to implement the MEC application support API for which the relevant sequence diagrams are described in clause 5.2.

7.1.2 Resource data types

7.1.2.1 Introduction

This clause defines data structures to be used in resource representations.

7.1.2.2 Type: TrafficRule

This type represents the general information of a traffic rule.

The attributes of the TrafficRule shall follow the indications provided in table 7.1.2.2-1.

Table 7.1.2.2-1: Attributes of TrafficRule

Attribute name	Data type	Cardinality	Description
trafficRuleId	String	1	Identify the traffic rule.
filterType	Enum (inlined)	1	Definition of filter per FLOW or PACKET. If FLOW, the filter matches upstream (e.g. UE→EPC) packets and downstream (e.g. EPC→UE) packets are handled in the same context.
priority	Int	1	Priority of this traffic rule within the range 0 to 255. If traffic rules conflict, the one with higher priority take precedence. See note 1.
trafficFilter	TrafficFilter	1..N	The filter used to identify specific packets that need to be handled by the MEC host.
action	Enum (inlined)	1	The action of the MEC host data plane when a packet matches the trafficFilter, the following actions are defined: • DROP • FORWARD_DECAPSULATED • FORWARD_ENCAPSULATED • PASSTHROUGH • DUPLICATE_DECAPSULATED • DUPLICATE_ENCAPSULATED
dstInterface	DestinationInterface	0..2	Describes the destination interface information. If the action is FORWARD_DECAPSULATED, FORWARD_ENCAPSULATED or PASSTHROUGH, one value shall be provided. If the action is DUPLICATE_DECAPSULATED or DUPLICATE_ENCAPSULATED, two values shall be provided. See note 2. If the action is DROP, no value shall be provided.
state	Enum (inlined)	1	Contains the traffic rule state: ACTIVE, INACTIVE. This attribute may be updated using HTTP PUT method.
NOTE 1: Value indicates the priority in descending order, i.e. with 0 as the highest priority and 255 as the lowest priority. NOTE 2: Some applications (like inline/tap) require two interfaces. The first interface in the case of inline/tap is on the client (e.g. UE) side and the second on the core network (e.g. EPC) side.			

7.1.2.3 Type: DnsRule

This type represents the general information of a DNS rule.

The attributes of the DnsRule shall follow the indications provided in table 7.1.2.3-1.

Table 7.1.2.3-1: Attributes of DnsRule

Attribute name	Data type	Cardinality	Description
dnsRuleId	String	1	Identifies the DNS Rule.
domainName	String	1	FQDN resolved by the DNS rule.
ipAddressType	Enum (inlined)	1	Specify the IP address type, value: IP_V6, IP_V4.
ipAddress	String	1	IP address associated with the FQDN resolved by the DNS rule.
ttl	Int	0..1	Time to live value, in seconds.
state	Enum (inlined)	1	Contains the DNS rule state: ACTIVE, INACTIVE. This attribute may be updated using HTTP PUT method.
NOTE: If no ttl value is provided, the DnsRule shall not expire.			

7.1.2.4 Type: TimingCaps

This type represents the information provided by the MEC platform in response to the "Timing capabilities Query" message.

The attributes of the TimingCaps shall follow the indications provided in table 7.1.2.4-1.

Table 7.1.2.4-1: Attributes of TimingCaps

Attribute name	Data type	Cardinality	Description
timeStamp	Structure (inlined)	0..1	
>seconds	Uint32	1	The seconds part of the Time. Time is defined as Unix-time since January 1, 1970, 00:00:00 UTC.
>nanoSeconds	Uint32	1	The nanoseconds part of the Time. Time is defined as Unix-time since January 1, 1970, 00:00:00 UTC.
ntpServers	Structure (inlined)	0..N	Number of available NTP servers.
>ntpServerAddrType	Enum (inlined)	1	Address type of NTP server with the following permitted values: • IP_ADDRESS • DNS_NAME
>ntpServerAddr	String	1	NTP server address.
>minPollingInterval	Uint32	1	Minimum poll interval for NTP messages, in seconds as a power of two. Range: 3 to 17.
>maxPollingInterval	Uint32	1	Maximum poll interval for NTP messages, in seconds as a power of two. Range: 3 to 17.
>localPriority	Uint32	1	NTP server local priority.
>authenticationOption	Enum (inlined)	1	NTP authentication option with the following permitted values: • NONE • SYMMETRIC_KEY • AUTO_KEY
>authenticationKeyNum	Uint32	0..1	Authentication key number. This configuration is valid and shall be present if authenticationOption is set to SYMMETRIC_KEY.
ptpMasters	Structure (inlined)	0..N	Number of available PTP Servers (referred to as "masters" in IEEE 1588™-2019 [i.2]).
>ptpMasterIpAddress	String	1	PTP Server (referred to as "master" in IEEE 1588™-2019 [i.2]) IP Address.
>ptpMasterLocalPriority	Uint32	1	PTP Server (referred to as "master" in IEEE 1588™-2019 [i.2]) local priority.
>delayReqMaxRate	Uint32	1	Acceptable maximum rate of the Delay_Req messages in packets per second.

7.1.2.5 Type: CurrentTime

This type represents the information provided by the MEC platform in response to the "Get Platform Time Request" message.

The attributes of the CurrentTime shall follow the indications provided in table 7.1.2.5-1.

Table 7.1.2.5-1: Attributes of CurrentTime

Attribute name	Data type	Cardinality	Description
seconds	Uint32	1	The seconds part of the Time. Time is defined as Unix-time since January 1, 1970, 00:00:00 UTC.
nanoSeconds	Uint32	1	The nanoseconds part of the Time. Time is defined as Unix-time since January 1, 1970, 00:00:00 UTC.
timeSourceStatus	Enum (inlined)	1	Platform Time Source status with the following permitted values: • TRACEABLE - time source is locked to the UTC time source • NONTRACEABLE - time source is not locked to the UTC time source

7.1.2.6 Type: AppInfo

This type represents the information provided by the MEC application instance as part of the "application registration request" and "application registration update" messages.

The attributes of the AppInfo shall follow the indications provided in table 7.1.2.6-1.

Table 7.1.2.6-1: Attributes of AppInfo

Attribute name	Data type	Cardinality	Description
appName	String	1	Name of the application. It shall be consistent with the appName in the AppD, if an AppD is available.
appProvider	String	0..1	Provider of the application. It shall be consistent with the appProvider in the AppD, if an AppD is available. See note 1.
appCategory	CategoryRef	0..1	Category of the application.
appDId	String	0..1	The application descriptor identifier. It is managed by the application provider to identify the application descriptor in a globally unique way. Shall be present if the application instance is instantiated by the MEC Management.
appInstanceId	String	0..1	Identifier of the application instance. Shall be present if the application instance is instantiated by the MEC Management.
endpoint	EndPointInfo	0..1	Endpoint information (e.g. URI, FQDN, IP address) of the application server, which is part of the application functionalities. Shall be present when isInsByMec is FALSE. See note 2.
appServiceRequired	ServiceDependency	0..N	Describes services a MEC application requires to run. ServiceDependency is defined in ETSI GS MEC 010-2 [4]. It shall not be provided if an AppD is available.
appServiceOptional	ServiceDependency	0..N	Describes services a MEC application may use if available. ServiceDependency is defined in ETSI GS MEC 010-2 [4]. It shall not be provided if an AppD is available.
appFeatureRequired	FeatureDependency	0..N	Describes features a MEC application requires to run. FeatureDependency is defined in ETSI GS MEC 010-2 [4]. It shall not be provided if an AppD is available.
appFeatureOptional	FeatureDependency	0..N	Describes features a MEC application may use if available. FeatureDependency is defined in ETSI GS MEC 010-2 [4]. It shall not be provided if an AppD is available.
scheds	array(ScheduledCommunicationTime)	0..N	The availability schedule of the application server. ScheduledCommunicationTime is defined in ETSI TS 129 122 [i.13]. See note 3.
svcArea	ServiceArea	0..1	The list of geographical and topological areas that the application server serves. ServiceArea is defined in ETSI TS 129 558 [i.11]. See note 3.
svcKpi	EASServiceKPI	0..1	Service characteristics provided by the application server. More information can be found in ETSI TS 123 558 [i.12], clause 8.2.5. EASServiceKPI is defined in ETSI TS 129 558 [i.11]. See note 3.
permLvl	array(String)	0..N	Level of service permissions supported by the application server. See note 3.
isInsByMec	Boolean	0..1	Indicate whether the application instance is instantiated by the MEC Management. Default to FALSE if absent.

Attribute name	Data type	Cardinality	Description
appProfile	AppProfile	0..1	Can be mapped to EAS profile as defined in ETSI TS 129 558 [i.11]. More information can be found in Annex C. See notes 1 and 2.
NOTE 1: If appProfile is present, appProvider shall be consistent with provId provided in EAS profile data type, i.e. the same. NOTE 2: If appProfile is present, endpoint shall refer to the same end point as endPt provided in EAS profile data type. NOTE 3: If appProfile is present, this attribute shall be consistent with the one provided in EAS profile data type, i.e. the same.			

7.1.3 Subscription data types

7.1.3.1 Introduction

This clause defines data structures that define criteria to be used in subscriptions.

7.1.3.2 Type: AppTerminationNotificationSubscription

This type represents a subscription to the notifications from the MEC platform related to MEC application instance termination/stop.

The attributes of the AppTerminationNotificationSubscription shall follow the indications provided in table 7.1.3.2-1.

Table 7.1.3.2-1: Attributes of AppTerminationNotificationSubscription

Attribute name	Data type	Cardinality	Description
subscriptionType	String	1	Shall be set to "AppTerminationNotificationSubscription".
callbackReference	Uri	1	URI selected by the MEC application instance to receive notifications on the subscribed MEC application instance management information. This shall be included in both the request and the response.
_links	Structure (inlined)	0..1	Object containing hyperlinks related to the resource. This shall only be included in the HTTP responses.
>self	LinkType	1	Self-referring URI.
appInstanceId	String	1	It is used as the filtering criterion for the subscribed events.

7.1.4 Notification data types

7.1.4.1 Introduction

This clause defines data structures that define notifications.

7.1.4.2 Type: AppTerminationNotification

This type represents the information that the MEC platform notifies the subscribed application instance about the corresponding application instance termination/stop.

The attributes of the AppTerminationNotification shall follow the indications provided in table 7.1.4.2-1.

Table 7.1.4.2-1: Attributes of AppTerminationNotification

Attribute name	Data type	Cardinality	Description
notificationType	String	1	Shall be set to "AppTerminationNotification".
operationAction	Enum (inlined)	1	Operation that is being performed on the MEC application instance: • STOPPING • TERMINATING
maxGracefulTimeout	Uint32	1	Maximum non-zero timeout value in seconds for graceful termination or graceful stop of an application instance.
_links	Structure (inlined)	1	Object containing hyperlinks related to the resource.
>subscription	LinkType	1	A link to the related subscription.
>confirmTermination	LinkType	0..1	Link to the task resource where to confirm termination/stop in case the application is ready to be terminated, or to be considered stopped by the MEC Platform, before expiry of the timeout.

7.1.4.3 Type: AppTerminationConfirmation

This type represents the information that the MEC application instance provides to the MEC platform when informing it that the application has completed its application level related terminate/stop actions, e.g. retention of application state in the case of stop.

The attributes of the AppTerminationConfirmation type shall follow the indications provided in table 7.1.4.3-1.

Table 7.1.4.3-1: Attributes of AppTerminationConfirmation

Attribute name	Data type	Cardinality	Description
operationAction	Enum (inlined)	1	Operation that is being performed on the MEC application instance: • STOPPING • TERMINATING The value shall match that sent in the corresponding AppTerminationNotification.

7.1.4.4 Type: AppReadyConfirmation

This type represents the information that the MEC application instance indicates to the MEC platform that it is up and running.

The attributes of the AppReadyConfirmation type shall follow the indications provided in table 7.1.4.4-1.

Table 7.1.4.4-1: Attributes of AppReadyConfirmation

Attribute name	Data type	Cardinality	Description
indication	String	1	Indication about the MEC application instance: • READY

7.1.5 Referenced structured data types

7.1.5.1 Introduction

This clause defines data structures that may be referenced from data structures defined in clauses 7.1.2 to 7.1.4, but are neither resource representations nor notifications.

7.1.5.2 Type: TrafficFilter

This type represents the traffic filter.

The attributes of the TrafficFilter shall follow the indications provided in table 7.1.5.2-1.

Table 7.1.5.2-1: Attributes of TrafficFilter

Attribute name	Data type	Cardinality	Description
srcAddress	String	0..N	An IP address or a range of IP address. For Ipv4, the IP address could be an IP address plus mask, or an individual IP address, or a range of IP addresses. For Ipv6, the IP address could be an IP prefix, or a range of IP prefixes.
dstAddress	String	0..N	An IP address or a range of IP address. For Ipv4, the IP address could be an IP address plus mask, or an individual IP address, or a range of IP addresses. For Ipv6, the IP address could be an IP prefix, or a range of IP prefixes.
srcPort	String	0..N	A port or a range of ports.
dstPort	String	0..N	A port or a range of ports.
protocol	String	0..N	Specify the protocol of the traffic filter.
tag	String	0..N	Used for tag based traffic rule.
srcTunnelAddress	String	0..N	Used for GTP tunnel based traffic rule.
tgtTunnelAddress	String	0..N	Used for GTP tunnel based traffic rule.
srcTunnelPort	String	0..N	Used for GTP tunnel based traffic rule.
dstTunnelPort	String	0..N	Used for GTP tunnel based traffic rule.
qCI	Int	0..1	Used to match all packets that have the same QCI.
dSCP	Int	0..1	Used to match all Ipv4 packets that have the same DSCP.
tC	Int	0..1	Used to match all Ipv6 packets that have the same TC.

7.1.5.3 Type: DestinationInterface

This type represents the destination interface.

The attributes of the DestinationInterface shall follow the indications provided in table 7.1.5.3-1.

Table 7.1.5.3-1: Attributes of DestinationInterface

Attribute name	Data type	Cardinality	Description
interfaceType	Enum (inlined)	1	Type of the interface, e.g. TUNNEL, MAC, IP, etc.
tunnelInfo	TunnelInfo	0..1	Included only if the destination interface type is "tunnel".
srcMacAddress	String	0..1	If the interface type is "MAC", source address identifies the MAC address of the interface.
dstMacAddress	String	0..1	If the interface type is "MAC", destination address identifies the MAC address of the interface. Only used for dstInterface.
dstIpAddress	String	0..1	If the interface type is "IP", destination address identifies the IP address of the remote destination. Only used for dstInterface.

7.1.5.4 Type: TunnelInfo

This type represents the tunnel information.

The attributes of the TunnelInfo shall follow the indications provided in table 7.1.5.4-1.

Table 7.1.5.4-1: Attributes of TunnelInfo

Attribute name	Data type	Cardinality	Description
tunnelType	Enum (inlined)	1	Type of the tunnel, e.g. GTP_U, GRE, etc.
tunnelDstAddress	String	0..1	Destination address of the tunnel.
tunnelSrcAddress	String	0..1	Source address of the tunnel.

7.1.6 Referenced simple data types and enumerations

Neither simple data types nor enumerations are defined for this API.

7.2 API definition

7.2.1 Introduction

This clause defines the resources and operations of the MEC application support API.

7.2.2 Global definitions and resource structure

All resource URIs of this API shall have the following root:

- {apiRoot}/{apiName}/{apiVersion}/

The "apiRoot" includes the scheme ("https"), host and optional port, and an optional prefix string. The "apiName" shall be set to "mec_app_support" and the "apiVersion" shall be set to "v2" for the present document. The "apiRoot", "apiName" and "apiVersion" can be discovered using the service registry. All resource URIs in clauses 7.2.3 to 7.2.14 are defined relative to the above root URI.

The API shall support HTTP over TLS as defined in clause 6.22 of ETSI GS MEC 009 [5].

The content format JSON shall be supported.

The JSON format shall be signalled by the content type "application/json".

This API shall use Oauth 2.0, as defined in clause 6.16 of ETSI GS MEC 009 [5]. This Oauth 2.0 authorization procedure shall occur only on TLS-protected connections.

This API supports additional application-related error information to be provided in the HTTP response when an error occurs. See clause 6.15 of ETSI GS MEC 009 [5] for more information.

Figure 7.2.2-1 illustrates the resource URI structure of this API.

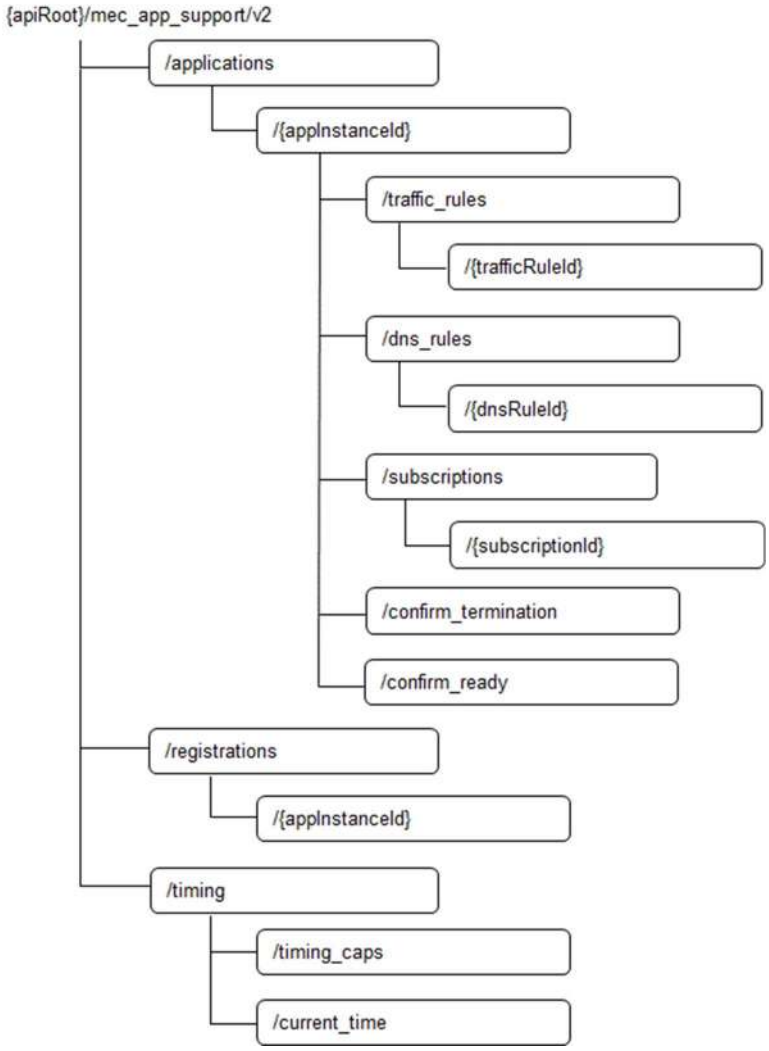


Figure 7.2.2-1: Resource URI structure of the MEC application support API

Table 7.2.2-1 provides an overview of the resources defined by the present specification for the MEC application support API, and the applicable HTTP methods.

NOTE: For any application instance that is identified by a particular appInstanceId value, there may be an entry in the "applications" resource or in the "registrations" resource only or in both. The appInstanceId is exposed by the MEP and managed jointly by the MEP and the MEPM.

Table 7.2.2-1: Resources and methods overview

Resource name	Resource URI	HTTP method	Meaning
Parent resource of all mecAppSupportSubscription of a subscriber	/applications/{appInstanceId}/subscriptions	GET	Retrieve information about a list of mecAppSupportSubscription resources for this subscriber.
		POST	Create a mecAppSupportSubscription resource.
Individual mecAppSupportSubscription	/applications/{appInstanceId}/subscriptions/{subscriptionId}	GET	Retrieve information about a mecAppSupportSubscription resource for this subscriber.
		DELETE	Delete a mecAppSupportSubscription resource.

Resource name	Resource URI	HTTP method	Meaning
Notification callback	Client provided callback reference	POST	Send a notification.
Parent resource of all mecTrafficRule of an application instance	/applications/{appInstanceId}/traffic_rules	GET	Retrieve information about a list of mecTrafficRule resources for an application instance.
Individual mecTrafficRule	/applications/{appInstanceId}/ traffic_rules/{trafficRuleId}	GET	Retrieve information about a mecTrafficRule resource.
		PUT	Update the information about a mecTrafficRule resource.
Parent resource of all mecDnsRule of an application instance	/applications/{appInstanceId}/dns_rules	GET	Retrieve information about a list of mecDnsRule resources for an application instance.
Individual mecDnsRule	/applications/{appInstanceId}/ dns_rules/{dnsRuleId}	GET	Retrieve information about a mecDnsRule resource.
		PUT	Update the information about a mecDnsRule resource.
Confirm termination task	/applications/{appInstanceId}/confirm_termination	POST	Confirm the application level termination of an App instance.
Confirm ready task	/applications/{appInstanceId}/confirm_ready	POST	Confirm the application instance is up and running.
MEC application instance registration	/registrations	POST	Register the MEC application instance to the MEC platform.
Existing MEC application instance registration	/registrations/{appInstanceId}	GET	Retrieve information about the MEC application instance registration to the MEC platform.
		PUT	Update the existing registration of that MEC application instance to the MEC platform.
		DELETE	Request deregistration of the application instance from the MEC platform.
mecTimingCaps	/timing/timing_caps	GET	Retrieve information about the mecTimingCaps resource.
mecCurrentTime	/timing/current_time	GET	Retrieve information about the mecCurrentTime resource.

7.2.3 Resource: all mecAppSupportSubscription

7.2.3.1 Description

This resource is used to represent all subscriptions of a subscriber to the notifications from the MEC platform.

7.2.3.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/subscriptions

Resource URI variables for this resource are defined in table 7.2.3.2-1.

Table 7.2.3.2-1: Resource URI variables for resource "all mecAppSupportSubscription"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance. Note that the appInstanceId is allocated by the MEC platform manager.

7.2.3.3 Resource methods

7.2.3.3.1 GET

The GET method may be used to request information about all subscriptions for this requestor. Upon success, the response contains message content with all the subscriptions for the requestor.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.3.3.1-1 and 7.2.3.3.1-2.

Table 7.2.3.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.3.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	SubscriptionLinkList	1	200 OK	Upon success, a response message content containing the list of links to the requested subscriptions is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.3.3.2 PUT

Not supported.

7.2.3.3.3 PATCH

Not supported.

7.2.3.3.4 POST

The POST method may be used to create a new subscription. One example use case is to create a new subscription to the MEC application termination notifications. Upon success, the response contains message content describing the created subscription.

POST HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.3.3.4-1 and 7.2.3.3.4-2.

Table 7.2.3.3.4-1: URI query parameters supported by the POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.3.3.4-2: Data structures supported by the POST request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	AppTerminationNotificationSubscription	1	Message content in the request contains a subscription to the MEC application termination notifications that is to be created.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	AppTerminationNotificationSubscription	1	201 Created	Upon success, the HTTP response shall include a "Location" HTTP header that contains the resource URI of the created subscription resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.3.3.5 DELETE

Not supported.

7.2.4 Resource: individual mecAppSupportSubscription

7.2.4.1 Description

This resource is used to represent a subscription to the notifications from the MEC platform. When this resource represents a subscription to the notifications related to MEC application instance termination/stop, it shall follow the data type of "AppTerminationNotificationSubscription" as specified in clause 7.1.3.2. The notifications that are related to a AppTerminationNotificationSubscription shall follow the data type of "AppTerminationNotification" as specified in clause 7.1.4.2.

7.2.4.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/subscriptions/{subscriptionId}

Resource URI variables for this resource are defined in table 7.2.4.2-1.

Table 7.2.4.2-1: Resource URI variables for resource "individual mecAppSupportSubscription"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance. Note that the appInstanceId is allocated by the MEC platform manager.
subscriptionId	Represents a subscription to the notifications from the MEC platform.

7.2.4.3 Resource methods

7.2.4.3.1 GET

The GET method requests information about a subscription for this requestor. Upon success, the response contains message content with the subscription for the requestor.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.4.3.1-1 and 7.2.4.3.1-2.

Table 7.2.4.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.4.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	AppTerminationNotification Subscription	1	200 OK	Upon success, a response message content containing the requested subscription is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.4.3.2 PUT

Not supported.

7.2.4.3.3 PATCH

Not supported.

7.2.4.3.4 POST

Not supported.

7.2.4.3.5 DELETE

This method deletes a mecAppSupportSubscription. This method is typically used in "Unsubscribing from event notifications" procedure as described in clause 5.2.6.3. Figure 7.2.4.3.5-1 shows the example message flows using DELETE method.

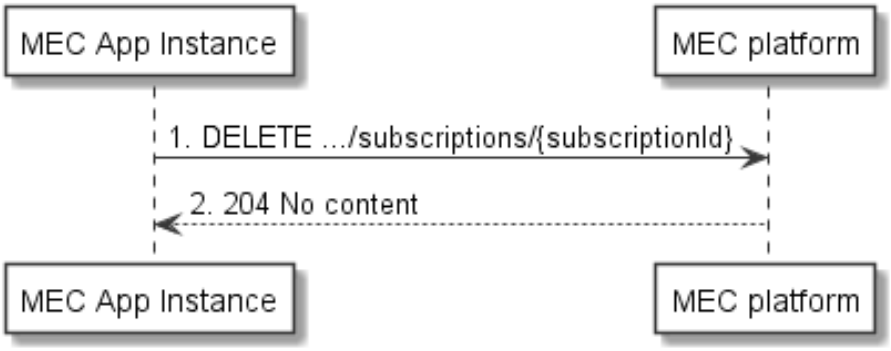


Figure 7.2.4.3.5-1: Unsubscribing from MEC application support event notifications

DELETE HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.4.3.5-1 and 7.2.4.3.5-2.

Table 7.2.4.3.5-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.4.3.5-2: Data structures supported by the DELETE request on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	n/a		204 No Content	
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.4a Resource: notification callback

7.2.4a.1 Description

This resource represents a notification callback, which can be used by the MEC Platform to send notifications related to application support events to a subscribed MEC application instance.

7.2.4a.2 Resource definition

The resource of callback URI is provided by the subscriber when subscribing to the notification.

7.2.4a.3 Resource methods

7.2.4a.3.1 GET

Not supported.

7.2.4a.3.2 PUT

Not supported.

7.2.4a.3.3 PATCH

Not supported.

7.2.4a.3.4 POST

The POST method delivers a notification from the MEC Platform to the subscriber.

This method shall follow the provisions specified in tables 7.2.4a.3.4-1 and 7.2.4a.3.4-2 for URI parameters, request and response data structures, and response codes.

Table 7.2.4a.3.4-1: URI query parameters supported by POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.4a.3.4-2: Data structures supported by POST request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	AppTermination Notification	1	A notification for application instance termination.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	n/a		204 No Content	The notification was delivered successfully. The response message content shall be empty.
	ProblemDetails	0..1	401 Unauthorized	It is used when the client did not submit credentials. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	429 Too Many Requests	It is used when a rate limiter has triggered. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

7.2.4a.3.5 DELETE

Not supported.

7.2.5 Resource: mecTimingCaps

7.2.5.1 Description

This resource is used to represent the timing capabilities of the MEC platform.

7.2.5.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/timing/timing_caps

Resource URI variables for this resource are defined in table 7.2.5.2-1.

Table 7.2.5.2-1: Resource URI variables for resource "mecTimingCaps"

Name	Definition
apiRoot	See clause 7.2.2

7.2.5.3 Resource methods

7.2.5.3.1 GET

This method retrieves the information of the platform's timing capabilities which corresponds to the timing capabilities query as described in clause 5.2.10.3. Figure 7.2.5.3.1-1 shows the example message flow for retrieving timing capabilities using GET method.



Figure 7.2.5.3.1-1: GET timing capabilities flow

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.5.3.1-1 and 7.2.5.3.1-2.

Table 7.2.5.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.5.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	Data type	Cardinality	Response codes	Remarks
	TimingCaps	1	200 OK	It is used to indicate nonspecific success. The response message content contains a representation of the resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.5.3.2 PUT

Not supported.

7.2.5.3.3 PATCH

Not supported.

7.2.5.3.4 POST

Not supported.

7.2.5.3.5 DELETE

Not supported.

7.2.6 Resource: mecCurrentTime

7.2.6.1 Description

This resource is used to represent the current time of the MEC platform.

7.2.6.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/timing/current_time

Resource URI variables for this resource are defined in table 7.2.6.2-1.

Table 7.2.6.2-1: Resource URI variables for resource "mecCurrentTime"

Name	Definition
apiRoot	See clause 7.2.2.

7.2.6.3 Resource methods

7.2.6.3.1 GET

This method retrieves the information of the platform's current time which corresponds to the get platform time procedure as described in clause 5.2.10.2. Figure 7.2.6.3.1-1 shows message flow for retrieving current time using GET method.



Figure 7.2.6.3.1-1: GET platform time API flow

This method shall comply with the URI query parameters, request and response data structures, as specified in tables 7.2.6.3.1-1 and 7.2.6.3.1-2.

Table 7.2.6.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.6.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	Data type	Cardinality	Response Codes	Remarks
	CurrentTime	1	200 OK	It is used to indicate nonspecific success. The response message content contains a representation of the resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.6.3.2 PUT

Not supported.

7.2.6.3.3 PATCH

Not supported.

7.2.6.3.4 POST

Not supported.

7.2.6.3.5 DELETE

Not supported.

7.2.7 Resource: all mecTrafficRule

7.2.7.1 Description

This resource is used to represent all the traffic rules associated with a MEC application instance, which follows the resource data type of "TrafficRule" as specified in clause 7.1.2.2.

7.2.7.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/traffic_rules

Resource URI variables for this resource are defined in table 7.2.7.2-1.

Table 7.2.7.2-1: Resource URI variables for resource "all mecTrafficRule"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance. Note that the appInstanceId is allocated by the MEC platform manager.

7.2.7.3 Resource methods

7.2.7.3.1 GET

This method retrieves information about all the traffic rules associated with a MEC application instance.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.7.3.1-1 and 7.2.7.3.1-2.

Table 7.2.7.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.7.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	Data type	Cardinality	Response codes	Remarks
	TrafficRule	0..N	200 OK	Upon success, a response message content containing an array of the TrafficRules is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.7.3.2 PUT

Not supported.

7.2.7.3.3 PATCH

Not supported.

7.2.7.3.4 POST

Not supported.

7.2.7.3.5 DELETE

Not supported.

7.2.8 Resource: individual mecTrafficRule

7.2.8.1 Description

This resource is used to represent a traffic rule, which follows the resource data type of "TrafficRule" as specified in clause 7.1.2.2.

7.2.8.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/traffic_rules/{trafficRuleId}

Resource URI variables for this resource are defined in table 7.2.8.2-1.

Table 7.2.8.2-1: Resource URI variables for resource "individual mecTrafficRule"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance. Note that the appInstanceId is allocated by the MEC platform manager.
trafficRuleId	Represents a traffic rule.

7.2.8.3 Resource methods

7.2.8.3.1 GET

This method retrieves information about a traffic rule associated with a MEC application instance.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.8.3.1-1 and 7.2.8.3.1-2.

Table 7.2.8.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.8.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	TrafficRule	1	200 OK	Upon success, a response message content containing the TrafficRules is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.8.3.2 PUT

This method activates, de-activates or updates a traffic rule. Figure 7.2.8.3.2-1 shows the message flow of "Traffic rule activation/deactivation/update" using PUT.

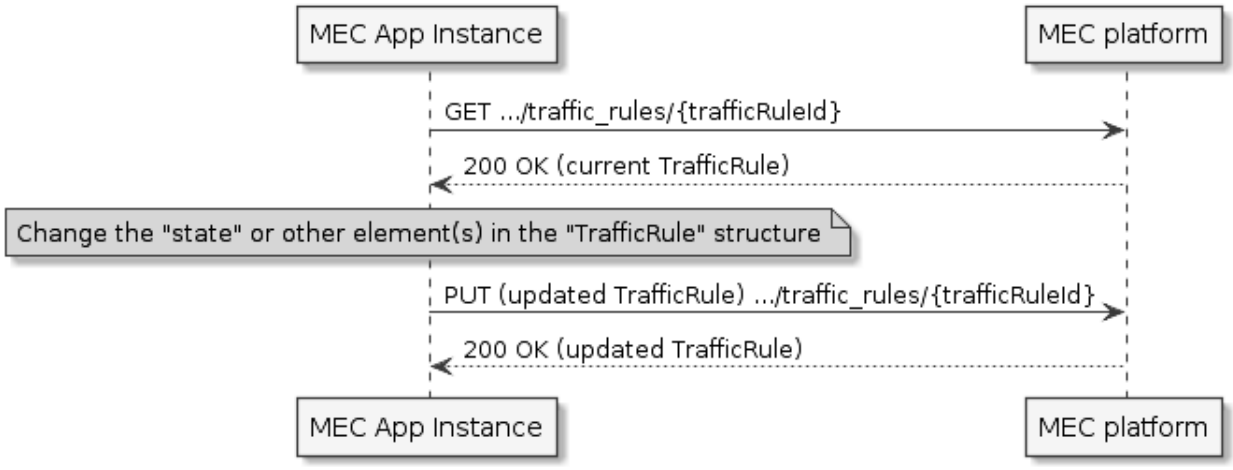


Figure 7.2.8.3.2-1: Traffic rule activation/deactivation/update using PUT

PUT HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.8.3.2-1 and 7.2.8.3.2-2.

Table 7.2.8.3.2-1: URI query parameters supported by the PUT method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.8.3.2-2: Data structures supported by the PUT request/response on this resource

	Data type	Cardinality	Response codes	Remarks
Request message content	TrafficRule	1		One or more updated attributes that are allowed to be changed (i.e. "state" or other attributes based on definition in clause 7.1.2.2) are included in the TrafficRule data structure in the message content of the request.
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	TrafficRule	1	200 OK	Upon success, a response message content containing data type describing the updated TrafficRule is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	412 Precondition Failed	It is used when a condition has failed during conditional requests, e.g. when using Etags to avoid write conflicts. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

7.2.8.3.3 PATCH

Not supported.

7.2.8.3.4 POST

Not supported.

7.2.8.3.5 DELETE

Not supported.

7.2.9 Resource: all mecDnsRule

7.2.9.1 Description

This resource is used to represent all the DNS rules associated with a MEC application instance, which follows the resource data type of "DnsRule" as specified in clause 7.1.2.3.

7.2.9.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/dns_rules

Resource URI variables for this resource are defined in table 7.2.9.2-1.

Table 7.2.9.2-1: Resource URI variables for resource "all mecDnsRule"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance. Note that the appInstanceId is allocated by the MEC platform manager.

7.2.9.3 Resource methods

7.2.9.3.1 GET

This method retrieves information about all the DNS rules associated with a MEC application instance.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.9.3.1-1 and 7.2.9.3.1-2.

Table 7.2.9.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.9.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	DnsRule	0..N	200 OK	Upon success, a response message content containing an array of the DnsRules is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.9.3.2 PUT

Not supported.

7.2.9.3.3 PATCH

Not supported.

7.2.9.3.4 POST

Not supported.

7.2.9.3.5 DELETE

Not supported.

7.2.10 Resource: individual mecDnsRule

7.2.10.1 Description

This resource is used to represent a DNS rule, which follows the resource data type of "DnsRule" as specified in clause 7.1.2.3.

7.2.10.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/dns_rules/{dnsRuleId}

Resource URI variables for this resource are defined in table 7.2.10.2-1.

Table 7.2.10.2-1: Resource URI variables for resource "individual mecDnsRule"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance. Note the appInstanceId is allocated by the MEC platform manager.
dnsRuleId	Represents a DNS rule.

7.2.10.3 Resource methods

7.2.10.3.1 GET

This method retrieves information about a DNS rule associated with a MEC application instance.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.10.3.1-1 and 7.2.10.3.1-2.

Table 7.2.10.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.10.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	DnsRule	1	200 OK	Upon success, a response message content containing the DnsRules is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.10.3.2 PUT

This method activates, de-activates or updates a DNS rule. Figure 7.2.10.3.2-1 shows the message flow of "DNS rule activation/deactivation" using PUT.

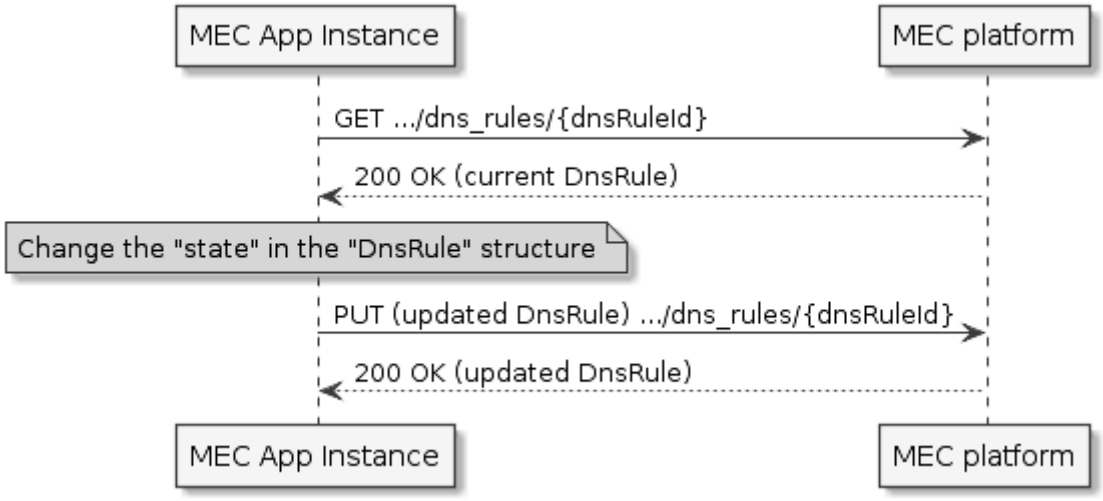


Figure 7.2.10.3.2-1: DNS rule activation/deactivation using PUT

PUT HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.10.3.2-1 and 7.2.10.3.2-2.

Table 7.2.10.3.2-1: URI query parameters supported by the PUT method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.10.3.2-2: Data structures supported by the PUT request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	DnsRule	1	The updated "state" is included in the message content of the request.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	DnsRule	1	200 OK	Upon success, a response message content containing data type describing the updated DnsRule is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	412 Precondition Failed	It is used when a condition has failed during conditional requests, e.g. when using Etags to avoid write conflicts. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

7.2.10.3.3 PATCH

Not supported.

7.2.10.3.4 POST

Not supported.

7.2.10.3.5 DELETE

Not supported.

7.2.11 Resource: confirm termination task

7.2.11.1 Description

This task resource allows a MEC application instance to confirm towards the MEC platform that it has completed the application level termination.

7.2.11.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/confirm_termination

Resource URI variables for this resource are defined in table 7.2.11.2-1.

Table 7.2.11.2-1: Resource URI variables for resource "confirm termination task"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance.

7.2.11.3 Resource methods

7.2.11.3.1 GET

Not supported.

7.2.11.3.2 PUT

Not supported.

7.2.11.3.3 PATCH

Not supported.

7.2.11.3.4 POST

The high-level MEC application instance graceful termination/stop flow is introduced in clause 5.2.3, with the full detail provided in figure 7.2.11.3.4-1. In step 1 the MEC Platform notifies the MEC application instance that it is to be gracefully terminated/stopped. In step 2 the MEC application instance responds with a 204 No Content to acknowledge that it has received the terminate/stop notification. It can then execute application level terminate/stop related actions. In step 3, once such actions have been completed, the MEC application instance uses the POST method to confirm the application level termination of the MEC application instance. Finally, in step 4, the MEC Platform responds with a 204 No Content. This POST method shall support the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.11.3.4-1 and 7.2.11.3.4-2.

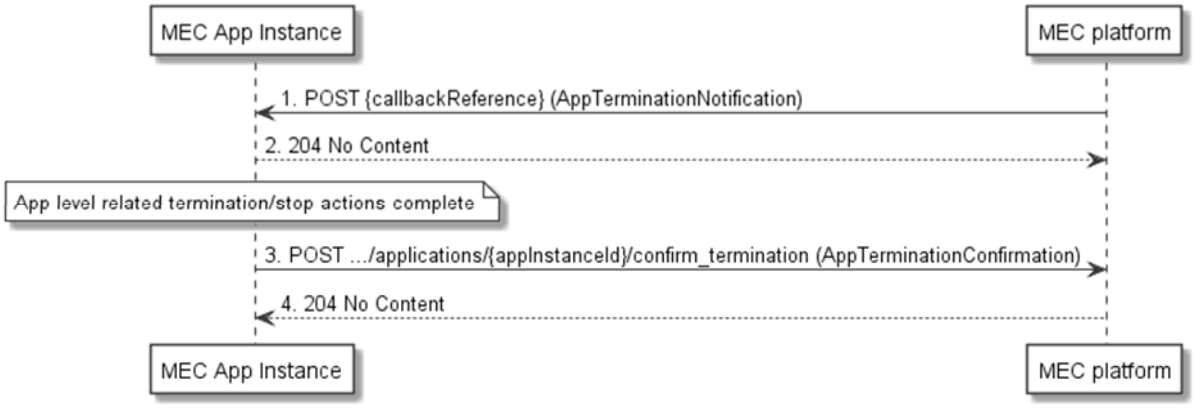


Figure 7.2.11.3.4-1: MEC application termination/stop notification and confirmation using POST

Table 7.2.11.3.4-1: URI query parameters supported by the POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.11.3.4-2: Data structures supported by the POST request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	AppTerminationConfirmation		Message content in the request contains the operational action the application instance is responding to.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	N/A		204 No Content	The request is acknowledged. The response message content shall be empty.
	ProblemDetails	0..1	401 Unauthorized	It is used when the client did not submit the appropriate credentials. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	409 Conflict	The operation cannot be executed currently, due to a conflict with the state of the resource. Typically, this is because the application instance resource is in NOT_INSTANTIATED state or because there is no termination ongoing. The response message content shall contain a ProblemDetails structure, in which the "detail" attribute shall convey more information about the error.
	ProblemDetails	0..1	429 Too Many Requests	It is used when a rate limiter has triggered. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

7.2.11.3.5 DELETE

Not supported.

7.2.12 Resource: confirm ready task

7.2.12.1 Description

This task resource allows a MEC application instance to confirm towards the MEC platform that it is up and running, which corresponds to step 4c described in clause 5.2.2.

7.2.12.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/applications/{appInstanceId}/confirm_ready

Resource URI variables for this resource are defined in table 7.2.12.2-1.

Table 7.2.12.2-1: Resource URI variables for resource "confirm ready task"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance.

7.2.12.3 Resource methods

7.2.12.3.1 GET

Not supported.

7.2.12.3.2 PUT

Not supported.

7.2.12.3.3 PATCH

Not supported.

7.2.12.3.4 POST

The POST method may be used by the MEC application instance to notify the MEC platform that it is up and running. POST HTTP method shall support the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.12.3.4-1 and 7.2.12.3.4-2.

Table 7.2.12.3.4-1: URI query parameters supported by the POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.12.3.4-2: Data structures supported by the POST request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	AppReadyConfirmation	1	Message content in the request contains the indication that the application instance is up and running.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	N/A		204 No Content	The request is acknowledged. The response message content shall be empty.
	ProblemDetails	0..1	401 Unauthorized	It is used when the client did not submit the appropriate credentials. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	409 Conflict	The operation cannot be executed currently, due to a conflict with the state of the resource. Typically, this is because the application instance resource is in NOT_INSTANTIATED state. The response message content shall contain a ProblemDetails structure, in which the "detail" attribute shall convey more information about the error.
	ProblemDetails	0..1	429 Too Many Requests	It is used when a rate limiter has triggered. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

7.2.12.3.5 DELETE

Not supported.

7.2.13 Resource: application instance registration

7.2.13.1 Description

This resource is used to represent registration of the MEC application instance to the MEC platform.

7.2.13.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/registrations

Resource URI variables for this resource are defined in table 7.2.13.2-1.

Table 7.2.13.2-1: Resource URI variables for resource "application instance registration"

Name	Definition
apiRoot	See clause 7.2.2.

7.2.13.3 Resource methods

7.2.13.3.1 GET

Not supported.

7.2.13.3.2 PUT

Not supported.

7.2.13.3.3 PATCH

Not supported.

7.2.13.3.4 POST

The POST method may be used by the MEC application instance to request its registration to the MEC platform. Upon success, the response contains message content describing the created registration.

POST HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.13.3.4-1 and 7.2.13.3.4-2.

Table 7.2.13.3.4-1: URI query parameters supported by the POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.13.3.4-2: Data structures supported by the POST request/response on this resource

	Data type	Cardinality	Remarks	
Request message content	AppInfo	1	The message content in the request contains the profile of the MEC application instance, calling the MEC platform to register the MEC application instance.	
Response message content	**Data type**	**Cardinality**	**Response Codes**	**Remarks**
	AppInfo	1	201 Created	Indicates successful resource creation for registration of the MEC application instance to the MEC platform. The resource URI shall be returned in the HTTP Location header field. appInstanceId shall be present in the newly created AppInfo resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	401 Unauthorized	It is used when the client did not submit credentials. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

7.2.13.3.5 DELETE

Not supported.

7.2.14 Resource: existing application instance registration

7.2.14.1 Description

This resource is used to represent the existing registration of the MEC application instance to the MEC platform.

7.2.14.2 Resource definition

Resource URI: {apiRoot}/mec_app_support/v2/registrations/{appInstanceId}

Resource URI variables for this resource are defined in table 7.2.14.2-1.

Table 7.2.14.2-1: Resource URI variables for resource "existing application instance registration"

Name	Definition
apiRoot	See clause 7.2.2.
appInstanceId	Represents a MEC application instance.

7.2.14.3 Resource methods

7.2.14.3.1 GET

The GET method may be used by the MEC application instance to retrieve information about the existing MEC application instance registration to the MEC platform. GET HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.14.3.1-1 and 7.2.14.3.1-2.

Table 7.2.14.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.14.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	AppInfo	1	200 OK	Upon success, a response message content containing the AppInfo is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

7.2.14.3.2 PUT

The PUT method may be used by the MEC application instance to update its registration to the MEC platform. PUT HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.14.3.2-1 and 7.2.14.3.2-2.

Table 7.2.14.3.2-1: URI query parameters supported by the PUT method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.14.3.2-2: Data structures supported by the PUT request/response on this resource

	Data type	Cardinality	Remarks	
Request message content	AppInfo	1	Message content in the request contains the profile of the application, calling the MEC platform to update the existing MEC application instance registration.	
Response message content	**Data type**	**Cardinality**	**Response Codes**	**Remarks**
	n/a		204 No Content	Upon success, a response 204 No Content without any response message content is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	401 Unauthorized	It is used when the client did not submit credentials. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

7.2.14.3.3 PATCH

Not supported.

7.2.14.3.4 POST

Not supported.

7.2.14.3.5 DELETE

The DELETE method is used to cancel the existing MEC application instance registration. Cancellation can be made by deleting the resource that represents the existing MEC application instance registration. DELETE HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 7.2.14.3.5-1 and 7.2.14.3.5-2.

Table 7.2.14.3.5-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 7.2.14.3.5-2: Data structures supported by the DELETE request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response Codes**	**Remarks**
	n/a		204 No Content	Upon success, a response 204 No Content without any response message content is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	401 Unauthorized	It is used when the client did not submit credentials. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

8 MEC service management API

8.1 Data model

8.1.1 Introduction

Clauses 8.1.2 to 8.1.6 specify the data types that are used to implement the MEC service management API for which the relevant sequence diagrams are described in clause 5.2.

8.1.2 Resource data types

8.1.2.1 Introduction

This clause defines data structures to be used in resource representations.

8.1.2.2 Type: ServiceInfo

This type represents the general information of a MEC service.

The attributes of the ServiceInfo shall follow the indications provided in table 8.1.2.2-1.

Table 8.1.2.2-1: Attributes of ServiceInfo

Attribute name	Data type	Cardinality	Description
serInstanceId	SerInstanceId	0..1	Identifier of the service instance assigned by the MEPM/MEC platform. For the uniqueness of the identifier across the MEC system, UUID format [i.7] is recommended. Shall be absent in POST requests, and present otherwise.
serName	SerName	1	The name of the service. This is how the service producing MEC application identifies the service instance it produces.
serCategory	CategoryRef	0..1	A Category reference. (The category resource is used to group product offerings, service and resource candidates in logical containers. Categories may contain other categories and/or product offerings, resource or service candidates.) (see note 1) For the "id" child attribute of the "serCategory" attribute, the example values include: 1. "RNI". 2. "Location". 3. "Bandwidth Management".
version	String	1	The version of the service.
state	ServiceState	1	Contains the service state.
transportId	String	0..1	Identifier of the platform-provided transport to be used by the service. Valid identifiers may be obtained using the "Transport information query" procedure. May be present in POST requests to signal the use of a platform-provided transport for the service, and shall be absent otherwise. See note 2.
transportInfo	TransportInfo	0..1	Information regarding the transport used by the service. May be present in POST requests to signal the use of an application-provided transport for the service, and shall be present otherwise. See note 2.
serializer	SerializerType	1	Indicate the supported serialization format of the service.
scopeOfLocality	LocalityType	0..1	The scope of locality as expressed by "consumedLocalOnly" and "isLocal". If absent, defaults to MEC_HOST. See notes 3, 5 and 6.
consumedLocalOnly	Boolean	0..1	Indicate whether the service can only be consumed by the MEC applications located in the same locality (as defined by scopeOfLocality) as this service instance (TRUE) or not (FALSE). Default to TRUE if absent.
isLocal	Boolean	0..1	Indicate whether the service is located in the same locality (as defined by scopeOfLocality) as the consuming MEC application (TRUE) or not (FALSE). Default to TRUE if absent. See note 4.
livenessInterval	Integer	0..1	Interval (in seconds) between two consecutive "heartbeat" messages (see clause 8.2.10.3.3). If the service-producing application supports sending "heartbeat" messages, it shall include this attribute in the registration request. In this case, the application shall either set the value of this attribute to zero or shall use this attribute to propose a non-zero positive value for the liveness interval. If the application has provided this attribute in the request and the MEC platform requires "heartbeat" messages, the MEC platform shall return this attribute value in the HTTP responses. The MEC platform may use the value proposed in the request or may choose a different value. If the MEC platform does not require "heartbeat" messages for this service instance it shall omit the attribute in responses.

Attribute name	Data type	Cardinality	Description
_links	Structure (inlined)	1	Links to resources related to this resource. Shall be absent in HTTP requests.
>self	LinkType	1	Link to this resource. Shall be present in HTTP responses.
>liveness	LinkType	0..1	Link to the "Individual mecServiceLiveness" resource where the MEC platform expects the service instance to send the liveness information. The structure of the URI of that resource is outside the scope of the present document. Shall be present in HTTP responses if the MEC platform requires "heartbeat" messages for this service instance and shall be absent otherwise.
NOTE 1: The service category may be included in the application descriptor. It may be allocated by the operator or by the application developer. NOTE 2: Either transportId or transportInfo but not both shall be present in POST requests. NOTE 3: Values NFVI_POP, ZONE and NFVI_NODE are used when the service instance is deployed as a VNF. NOTE 4: The isLocal is used only in service availability query response and service availability subscription/notification messages. NOTE 5: Value ZONE_GROUP can be used when the service instance is deployed as a VNF. NOTE 6: Regarding the value MEC_SYSTEM, if the service is running on the same MEC system as the MEC app, then it will be local to it.			

NOTE: In the present document it is not specified on service availability announcements outside a MEC system.

8.1.2.3 Type: TransportInfo

This type represents the transport information. The attributes of the TransportInfo type shall follow the indications provided in table 8.1.2.3-1.

Table 8.1.2.3-1: Attributes of TransportInfo

Attribute name	Data type	Cardinality	Description
id	String	1	The identifier of this transport.
name	String	1	The name of this transport.
description	String	0..1	Human-readable description of this transport.
type	TransportType	1	Type of the transport.
protocol	String	1	The name of the protocol used. Shall be set to "HTTP" for a REST API.
version	String	1	The version of the protocol used.
endpoint	EndPointInfo	1	Information about the endpoint to access the transport.
security	SecurityInfo	1	Information about the security used by the transport.
implSpecificInfo	Not specified	0..1	Additional implementation specific details of the transport.

8.1.2.4 Type: ServiceLivenessInfo

This type represents the liveness information of a MEC service instance. The attributes of the "ServiceLivenessInfo" type shall follow the indications provided in table 8.1.2.4-1.

Table 8.1.2.4-1: Attributes of ServiceLivenessInfo

Attribute name	Data type	Cardinality	Description
state	ServiceState	1	Liveness state of the MEC service instance. The valid values are defined in clause 8.1.6.6.
timeStamp	Structure (inlined)	1	The time when the last "heartbeat" message was received by MEC platform.
>seconds	Uint32	1	The seconds part of the time. Time is defined as Unix-time since January 1, 1970, 00:00:00 UTC.
>nanoSeconds	Uint32	1	The nanoseconds part of the time. Time is defined as Unix-time since January 1, 1970, 00:00:00 UTC.
interval	Integer	1	The interval (in seconds) between two consecutive "heartbeat" messages (see clause 8.2.10.3.3) that MEC platform has determined.

8.1.2.5 Type: ServiceLivenessUpdate

This type represents changes to the liveness information of a MEC service instance, following the syntax of JSON Merge Patch specified in IETF RFC 7386 [17]. The "ServiceLivenessUpdate" type contains the subset of the attributes of the "ServiceLivenessInfo" type which are allowed to be modified by the PATCH method.

The attributes of the "ServiceLivenessUpdate" type shall follow the indications provided in table 8.1.2.5-1.

Table 8.1.2.5-1: Attributes of ServiceLivenessUpdate

Attribute name	Data type	Cardinality	Description
state	ServiceState	1	Update to the state of the MEC service instance to indicate that the service is still alive ("heartbeat" message). Shall be set to "ACTIVE".

8.1.3 Subscription data types

8.1.3.1 Introduction

This clause defines data structures that define criteria to be used in subscriptions.

8.1.3.2 Type: SerAvailabilityNotificationSubscription

This type represents a subscription to the notifications from the MEC platform regarding the availability of a MEC service or a list of MEC services.

The attributes of the SerAvailabilityNotificationSubscription shall follow the indications provided in table 8.1.3.2-1.

Table 8.1.3.2-1: Attributes of SerAvailabilityNotificationSubscription

Attribute name	Data type	Cardinality	Description
subscriptionType	String	1	Shall be set to "SerAvailabilityNotificationSubscription".
callbackReference	Uri	1	URI selected by the MEC application instance to receive notifications on the subscribed MEC service availability information. This shall be included in both the request and the response.
_links	Structure (inlined)	0..1	Object containing hyperlinks related to the resource. This shall only be included in the HTTP responses.
>self	LinkType	1	Self-referring URI.
filteringCriteria	Structure (inlined)	0..1	Filtering criteria to match services for which events are requested to be reported. If absent, matches all services. All child attributes are combined with the logical "AND" operation.
>serInstanceIds	SerInstanceId	0..N	Identifiers of service instances about which to report events. See note.
>serNames	SerName	0..N	Names of services about which to report events. See note.
>serCategories	CategoryRef	0..N	Categories of services about which to report events. See note.
>states	ServiceState	0..N	States of the services about which to report events. If the event is a state change, this filter represents the state after the change.
>isLocal	Boolean	0..1	Restrict event reporting to whether the service is local to the MEC platform where the subscription is managed.
NOTE: The attributes "serInstanceIds", "serNames" and "serCategories" provide mutually-exclusive alternatives to define a set of services. Only one of them may be present.			

8.1.4 Notification data types

8.1.4.1 Introduction

This clause defines data structures that define notifications.

8.1.4.2 Type: ServiceAvailabilityNotification

This type represents the service availability information that is used in the following cases:

- when the MEC platform announces the newly available services to the authorized relevant MEC applications (e.g. the applications that indicate the services as "optional" or "required") that are subscribed to the corresponding service availability notifications;
- when the MEC platform notifies the authorized relevant applications that are subscribed to the corresponding service availability notifications about the service availability changes.

The attributes of the ServiceAvailabilityNotification shall follow the indications provided in table 8.1.4.2-1.

Table 8.1.4.2-1: Attributes of ServiceAvailabilityNotification

Attribute name	Data type	Cardinality	Description
notificationType	String	1	Shall be set to "SerAvailabilityNotification".
serviceReferences	Structure (inlined)	1..N	List of links to services whose availability has changed.
>link	LinkType	0..1	Link to the resource representing the individual service. Shall be present unless "changeType" = "REMOVED".
>serName	SerName	1	Name of the service
>serInstanceId	SerInstanceId	1	Identifier of the service
>state	ServiceState	1	State of the service after the modification.
>changeType	ServiceChange	1	Type of the change.
_links	Structure (inlined)	1	Object containing hyperlinks related to the resource.
>subscription	LinkType	1	A link to the related subscription.

8.1.5 Referenced structured data types

8.1.5.1 Introduction

This clause defines data structures that may be referenced from data structures defined in clauses 8.1.2 to 8.1.4, but may neither be resource representations nor notifications.

8.1.5.2 Type: CategoryRef

This type represents the category reference.

The attributes of the CategoryRef shall follow the indications provided in table 8.1.5.2-1.

Table 8.1.5.2-1: Attributes of CategoryRef

Attribute name	Data type	Cardinality	Description
href	Uri	1	Reference of the category.
id	String	1	Unique identifier of the category.
name	String	1	Name of the category.
version	String	1	Category version.

8.1.5.3 Type: EndPointInfo

This type represents information about a transport endpoint. The attributes of the EndPointInfo shall follow the indications provided in table 8.1.5.3-1.

Table 8.1.5.3-1: Attributes of EndPointInfo

Attribute name	Data type	Cardinality	Description
uris	String	0..N	Entry point information of the service as string, formatted according to URI syntax (see IETF RFC 3986 [8]). Shall be used for REST APIs. See note.
fqdn	String	0..N	Fully Qualified Domain Name of the service. See note.
addresses	Structure (inlined)	0..N	Entry point information of the service as one or more pairs of IP address and port. See note.
>host	String	1	Host portion of the address.
>port	Int	1	Port portion of the address.
alternative	Not specified	0..1	Entry point information of the service in a format defined by an implementation, or in an external specification. See note.
NOTE: Exactly one of "uris", "fqdn", "addresses" or "alternative" shall be present.			

8.1.5.4 Type: SecurityInfo

This type represents security information related to a transport.

NOTE: For the use of alternative transport mechanisms by implementations, or for their specification in future versions of the present document, it is foreseen that the "SecurityInfo" structure may contain additional attributes that allow the MEC application to discover the applicable security-related parameters of these mechanisms.

Table 8.1.5.4-1: Attributes of SecurityInfo

Attribute name	Data type	Cardinality	Description
oAuth2Info	Oauth2Info	0..1	Parameters related to use of Oauth 2.0. Shall be present in case Oauth 2.0 (see IETF RFC 6749 [13]) is supported to secure the provision of the service over the transport, as specified in clause 6.16 of ETSI GS MEC 009 [5].
(extensions)	Not specified	0..N	Extensions for alternative transport mechanisms. These extensions depend on the actual transport, and are out of scope of the present document. For instance, such extensions may be used to signal the necessary parameters for the client to use TLS-based authorization defined for alternative transports (see ETSI GS MEC 009 [5] for more information). The name of the extensions attributes can be chosen as defined by the actual transport.

8.1.5.5 Type: Oauth2Info

This type represents security information related to Oauth 2.0 (see IETF RFC 6749 [13]).

In the present document, only security information for the client credentials grant type of Oauth 2.0 is specified. All parameters related to Oauth 2.0, including additional attributes that might need to be added when more grant types are supported in the future, shall be contained in the "Oauth2Info" structure. For the support of the Oauth 2.0 client credentials grant type, the attributes of the "Oauth2Info" data type shall follow the indications provided in table 8.1.5.5-1.

Table 8.1.5.5-1: Attributes of Oauth2Info

Attribute name	Data type	Cardinality	Description
grantTypes	Enum (inlined)	1..4	List of supported Oauth 2.0 grant types. Each entry shall be one of the following permitted values: • OAUTH2_AUTHORIZATION_CODE (Authorization code grant type) • OAUTH2_IMPLICIT_GRANT (Implicit grant type) • OAUTH2_RESOURCE_OWNER (Resource owner password credentials grant type) • OAUTH2_CLIENT_CREDENTIALS (Client credentials grant type) Only the value "OAUTH2_CLIENT_CREDENTIALS" is supported in the present document.
tokenEndpoint	Uri	0..1	The token endpoint. It shall be present unless the grant type is OAUTH2_IMPLICIT_GRANT.

8.1.6 Referenced simple data types and enumerations

8.1.6.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in clauses 8.1.2 to 8.1.5.

8.1.6.2 Simple data types

The simple data type defined for this API are provided in table 8.1.6.2-1.

Table 8.1.6.2-1: Simple data types

Type name	Description
SerInstanceId	String representing the identifier of the service.
SerName	String representing the name of the service.

8.1.6.3 Enumeration: SerializerType

The enumeration SerializerType represents types of serializers. This enumeration shall be extensible. It shall comply with the provisions defined in table 8.1.6.3-1.

Table 8.1.6.3-1: Enumeration SerializerType

Enumeration value	Description
JSON	Javascript object notation [9].
XML	eXtensible Mark-up Language version 1.1 [10].
PROTOBUF3	Protocol buffers version 3 [i.3].
NOTE: The enumeration values above shall represent the serializers as defined by the referenced specifications.	

8.1.6.4 Enumeration: TransportType

The enumeration TransportType represents types of transports. It shall comply with the provisions defined in table 8.1.6.4-1. This enumeration shall be extensible.

Table 8.1.6.4-1: Enumeration TransportType

Enumeration value	Description
REST_HTTP	RESTful API using HTTP (as defined in IETF RFC 9110 [11]).
MB_TOPIC_BASED	Topic-based message bus which routes messages to receivers based on subscriptions, if a pattern passed on subscription matches the topic of the message. EXAMPLE: MQTT (see [i.4]).
MB_ROUTING	Routing-based message bus which routes messages to receivers based on subscriptions, if a key passed on subscription is equal to the key of the message.
MB_PUBSUB	Publish-subscribe based message bus which distributes messages to all subscribers.
RPC	Remote procedure call. EXAMPLE: GRPC (see [i.5]).
RPC_STREAMING	Remote procedure call supporting streams of requests and responses. EXAMPLE: GRPC (see [i.5]).
WEBSOCKET	Websockets as defined in IETF RFC 6455 [12].

8.1.6.5 Enumeration: LocalityType

The enumeration LocalityType represents types of locality. It shall comply with the provisions defined in table 8.1.6.5-1.

Table 8.1.6.5-1: Enumeration LocalityType

Enumeration value	Description
MEC_SYSTEM	MEC system.
MEC_HOST	MEC host.
NFVI_POP	NFVI PoP.
ZONE	Resource zone, as defined in ETSI GS NFV-IFA 007 [15].
ZONE_GROUP	Group of resource zones, as defined in ETSI GS NFV-IFA 007 [15].
NFVI_NODE	NFVI node.

NOTE: In the present document it is not specified on service availability announcements outside a MEC system.

8.1.6.6 Enumeration: ServiceState

The enumeration ServiceState represents possible states of a MEC service instance. This enumeration shall comply with the provisions defined in table 8.1.6.6-1.

Table 8.1.6.6-1: Enumeration ServiceState

Enumeration value	Description
ACTIVE	The service is active.
INACTIVE	The service is inactive.
SUSPENDED	The service is suspended because its producer did not send a "heartbeat" message in the expected time interval.

8.1.6.7 Enumeration: ServiceChange

The enumeration ServiceChange represents changes to a MEC service. This enumeration shall be extensible. It shall comply with the provisions defined in table 8.1.6.7-1.

Table 8.1.6.7-1: Enumeration ServiceChange

Enumeration value	Description
ADDED	The service was newly added.
REMOVED	The service was removed.
STATE_CHANGED	Only the state of the service was changed.
ATTRIBUTES_CHANGED	At least one attribute of the service other than state was changed. The change may or may not include changing the state.

8.2 API definition

8.2.1 Introduction

This clause defines the resources and operations of the MEC service management API.

8.2.2 Global definitions and resource structure

All resource URIs of this API shall have the following root:

- {apiRoot}/{apiName}/{apiVersion}/

The "apiRoot" includes the scheme ("https"), host and optional port, and an optional prefix string. The "apiName" shall be set to "mec_service_mgmt" and the "apiVersion" shall be set to "v1" for the present document. The "apiRoot", "apiName" and "apiVersion" can be discovered by means outside the scope of the present document. All resource URIs in clauses 8.2.3 to 8.2.10 are defined relative to the above root URI.

The API shall support HTTP over TLS as defined in clause 6.22 of ETSI GS MEC 009 [5].

The content format JSON shall be supported.

The JSON format shall be signalled by the content type "application/json".

This API shall use Oauth 2.0, as defined in clause 6.16 of ETSI GS MEC 009 [5]. This Oauth 2.0 authorization procedure shall occur only on TLS-protected connections.

This API supports additional application-related error information to be provided in the HTTP response when an error occurs. See clause 6.15 of ETSI GS MEC 009 [5] for more information. Figure 8.2.2-1 illustrates the resource URI structure of this API.

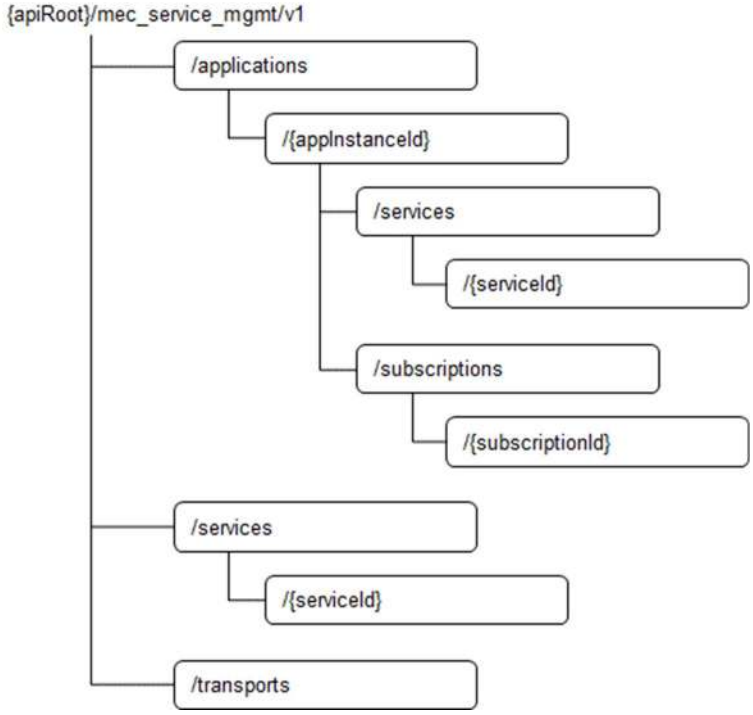


Figure 8.2.2-1: Resource URI structure of the MEC service management API

Table 8.2.2-1 provides an overview of the resources defined by the present specification for the MEC applications support API, and the applicable HTTP methods.

Table 8.2.2-1: Resources and methods overview

Resource name	Resource URI	HTTP method	Meaning
A list of mecService	/services	GET	Retrieve information about a list of mecService resources.
Individual mecService	/services/{serviceId}	GET	Retrieve information about a mecService resource.
A list of mecService of an application instance	/applications/{appInstanceId}/services	GET	Retrieve information about a list of mecService resources of an application instance.
		POST	Create a mecService resource of an application instance.
Individual mecService of an application instance	/applications/{appInstanceId}/services/{serviceId}	GET	Retrieve information about a mecService resource of an application instance.
		PUT	Update the information about a mecService resource of an application instance.
		DELETE	Delete a mecService resource.
Parent resource of all mecSrvMgmtSubscription of a subscriber	/applications/{appInstanceId}/subscriptions	GET	Retrieve information about a list of mecSrvMgmtSubscription resources for this subscriber.
		POST	Create a mecSrvMgmtSubscription resource.

Resource name	Resource URI	HTTP method	Meaning
Individual mecSrvMgmtSubscription	/applications/{appInstanceId}/subscriptions/{subscriptionId}	GET	Retrieve information about a mecSrvMgmtSubscription resource for this subscriber.
		DELETE	Delete a mecSrvMgmtSubscription resource.
Notification callback	Client provided callback reference	POST	Send a notification.
A list of mecTransport	/transports	GET	Retrieve information about the available transports.
Individual mecServiceLiveness	See note.	GET	Retrieve information about the liveness of a MEC service instance produced by an application instance.
		PATCH	Send a "heartbeat" message related to a MEC service instance.
NOTE: The URI of this resource is allocated by the MEC platform.			

8.2.3 Resource: a list of mecService

8.2.3.1 Description

This resource is used to represent a list of MEC service instances.

8.2.3.2 Resource definition

Resource URI: {apiRoot}/mec_service_mgmt/v1/services

Resource URI variables for this resource are defined in table 8.2.3.2-1.

Table 8.2.3.2-1: Resource URI variables for resource "a list of mecService"

Name	Definition
apiRoot	See clause 8.2.2.

8.2.3.3 Resource methods

8.2.3.3.1 GET

This method retrieves information about a list of mecService resources. This method is typically used in "service availability query" procedure as described in clause 5.2.5. Figure 8.2.3.3.1-1 shows the example message flows using GET method.

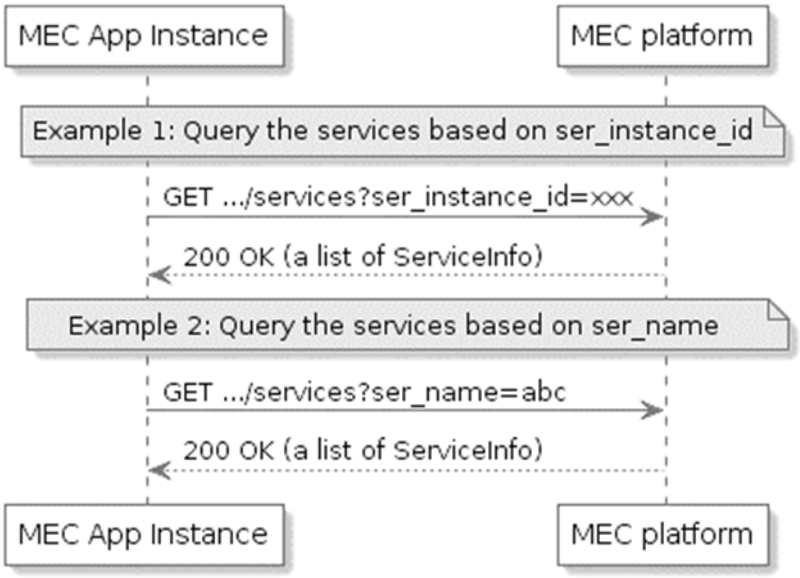


Figure 8.2.3.3.1-1: Service availability query

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.3.3.1-1 and 8.2.3.3.1-2. When no URI query parameter is present, all the relevant mecService resources to the requestor will be returned.

Table 8.2.3.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
ser_instance_id	String	0..N	A MEC application instance may use multiple ser_instance_ids as an input parameter to query the availability of a list of MEC service instances. Ser_instance_id corresponds to serInstanceId defined in table 8.1.2.2-1. See note.
ser_name	String	0..N	A MEC application instance may use multiple ser_names as an input parameter to query the availability of a list of MEC service instances. Ser_name corresponds to serName defined in table 8.1.2.2-1. See note.
ser_category_id	String	0..1	A MEC application instance may use ser_category_id as an input parameter to query the availability of a list of MEC service instances in a serCategory. Ser_category_id corresponds to serCategory defined in table 8.1.2.2-1. See note.
scope_of_locality	LocalityType	0..1	A MEC application instance may use scope_of_locality as an input parameter to query the availability of a list of MEC service instances with a certain scope of locality, as defined in LocalityType in table 8.1.6.5-1. Scope_of_locality corresponds to scopeOfLocality defined in table 8.1.2.2-1.
consumed_local_only	Boolean	0..1	A MEC application instance may use consumed_local_only as an input parameter to query the availability of a list of MEC service instances that can be consumed only locally. Consumed_local_only corresponds to consumedLocalOnly defined in table 8.1.2.2-1.
is_local	Boolean	0..1	A MEC application instance may use is_local as an input parameter to query the availability of a list of MEC service instances in the local MEC host or in local and remote MEC hosts. Is_local corresponds to isLocal defined in table 8.1.2.2-1.
NOTE: Either "ser_instance_id" or "ser_name" or "ser_category_id" or none of them shall be present.			

Table 8.2.3.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceInfo	0..N	200 OK	Upon success, a response message content containing an array of the mecServices is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	414 URI Too Long	It is used to indicate that the server is refusing to process the request because the request URI is longer than the server is willing or able to process.

8.2.3.3.2 PUT

Not supported.

8.2.3.3.3 PATCH

Not supported.

8.2.3.3.4 POST

Not supported.

8.2.3.3.5 DELETE

Not supported.

8.2.4 Resource: individual mecService

8.2.4.1 Description

This resource is used to represent a MEC service instance, which follows the resource data type of "ServiceInfo" as specified in clause 8.1.2.2.

8.2.4.2 Resource definition

Resource URI: {apiRoot}/mec_service_mgmt/v1/services/{serviceId}

Resource URI variables for this resource are defined in table 8.2.4.2-1.

Table 8.2.4.2-1: Resource URI variables for resource "individual mecService"

Name	Definition
apiRoot	See clause 8.2.2.
serviceId	Represents a MEC service instance (see note).
NOTE: serviceId corresponds to serInstanceId (defined in table 8.1.2.2-1).	

8.2.4.3 Resource methods

8.2.4.3.1 GET

This method retrieves information about a mecService resource. This method is typically used in "service availability query" procedure as described in clause 5.2.5.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.4.3.1-1 and 8.2.4.3.1-2.

Table 8.2.4.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.4.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceInfo	1	200 OK	It is used to indicate nonspecific success. The response message content contains a representation of the resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.4.3.2 PUT

Not supported.

8.2.4.3.3 PATCH

Not supported.

8.2.4.3.4 POST

Not supported.

8.2.4.3.5 DELETE

Not supported.

8.2.5 Resource: a list of mecTransport

8.2.5.1 Description

This resource is used to represent a list of transports provided by the MEC platform.

8.2.5.2 Resource definition

Resource URI: {apiRoot}/mec_service_mgmt/v1/transports

Resource URI variables for this resource are defined in table 8.2.5.2-1.

Table 8.2.5.2-1: Resource URI variables for resource "a list of mecTransport"

Name	Definition
apiRoot	See clause 8.2.2.

8.2.5.3 Resource methods

8.2.5.3.1 GET

This method retrieves information about a list of available transports. This method is typically used by a service-producing application to discover transports provided by the MEC platform in the "transport information query" procedure as described in clause 5.2.9. Figure 8.2.5.3.1-1 shows the example message flows using GET method.

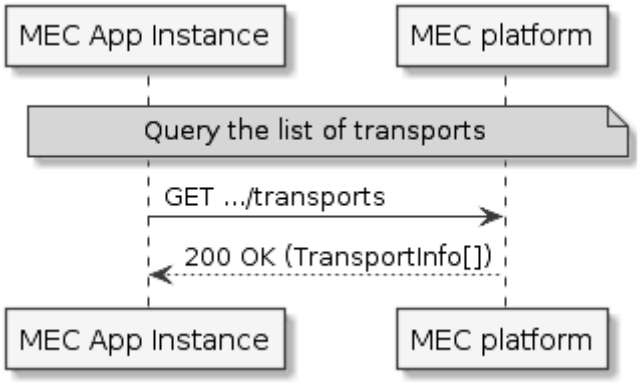


Figure 8.2.5.3.1-1: Transport information query

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.5.3.1-1 and 8.2.5.3.1-2.

Table 8.2.5.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.5.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	TransportInfo	0..N	200 OK	Upon success, a response message content containing an array describing the available transports is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.5.3.2 PUT

Not supported.

8.2.5.3.3 PATCH

Not supported.

8.2.5.3.4 POST

Not supported.

8.2.5.3.5 DELETE

Not supported.

8.2.6 Resource: a list of mecService of an application instance

8.2.6.1 Description

This resource is used to represent a list of MEC service instances that is associated with an application instance.

8.2.6.2 Resource definition

Resource URI: {apiRoot}/mec_service_mgmt/v1/applications/{appInstanceId}/services

Resource URI variables for this resource are defined in table 8.2.6.2-1.

Table 8.2.6.2-1: Resource URI variables for resource "a list of mecService of an application instance"

Name	Definition
apiRoot	See clause 8.2.2.
appInstanceId	Represents a MEC application instance.

8.2.6.3 Resource methods

8.2.6.3.1 GET

This method retrieves information about a list of mecService resources that is associated with an application instance. This method is typically used in "service availability query" procedure as described in clause 5.2.5. Figure 8.2.6.3.1-1 shows the example message flows using GET method.

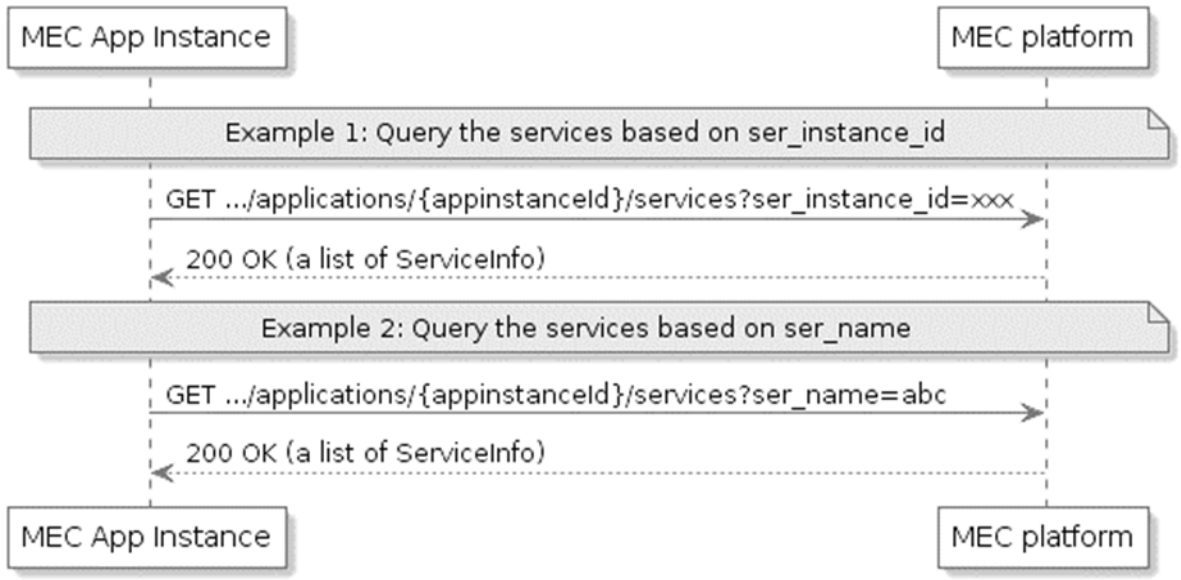


Figure 8.2.6.3.1-1: Service availability query

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.6.3.1-1 and 8.2.6.3.1-2. When no URI query parameter is present, all the relevant mecService resources to the requestor will be returned.

Table 8.2.6.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
ser_instance_id	String	0..N	A MEC application instance may use multiple ser_instance_ids as an input parameter to query the availability of a list of MEC service instances. Ser_instance_id corresponds to serInstanceId defined in table 8.1.2.2-1. See note.
ser_name	String	0..N	A MEC application instance may use multiple ser_names as an input parameter to query the availability of a list of MEC service instances. Ser_name corresponds to serName defined in table 8.1.2.2-1. See note.
ser_category_id	String	0..1	A MEC application instance may use ser_category_id as an input parameter to query the availability of a list of MEC service instances in a serCategory. Ser_category_id corresponds to serCategory defined in table 8.1.2.2-1. See note.
scope_of_locality	LocalityType	0..1	A MEC application instance may use scope_of_locality as an input parameter to query the availability of a list of MEC service instances with a certain scope of locality, as defined in LocalityType in table 8.1.6.5-1. Scope_of_locality corresponds to scopeOfLocality defined in table 8.1.2.2-1.
consumed_local_only	Boolean	0..1	A MEC application instance may use consumed_local_only as an input parameter to query the availability of a list of MEC service instances that can be consumed only locally. Consumed_local_only corresponds to consumedLocalOnly defined in table 8.1.2.2-1.

Name	Data type	Cardinality	Remarks
is_local	Boolean	0..1	A MEC application instance may use is_local as an input parameter to query the availability of a list of MEC service instances in the local MEC host or in local and remote MEC hosts. Is_local corresponds to isLocal defined in table 8.1.2.2-1.
NOTE: Either "ser_instance_id" or "ser_name" or "ser_category_id" or none of them shall be present.			

Table 8.2.6.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceInfo	0..N	200 OK	Upon success, a response message content containing an array of the mecServices is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	414 URI Too Long	It is used to indicate that the server is refusing to process the request because the request URI is longer than the server is willing or able to process.

8.2.6.3.2 PUT

Not supported.

8.2.6.3.3 PATCH

Not supported.

8.2.6.3.4 POST

This method is used to create a mecService resource that is associated with the application instance. This method is typically used in "service availability update and new service registration" procedure as described in clause 5.2.4. Figure 8.2.6.3.4-1 shows the message flow.

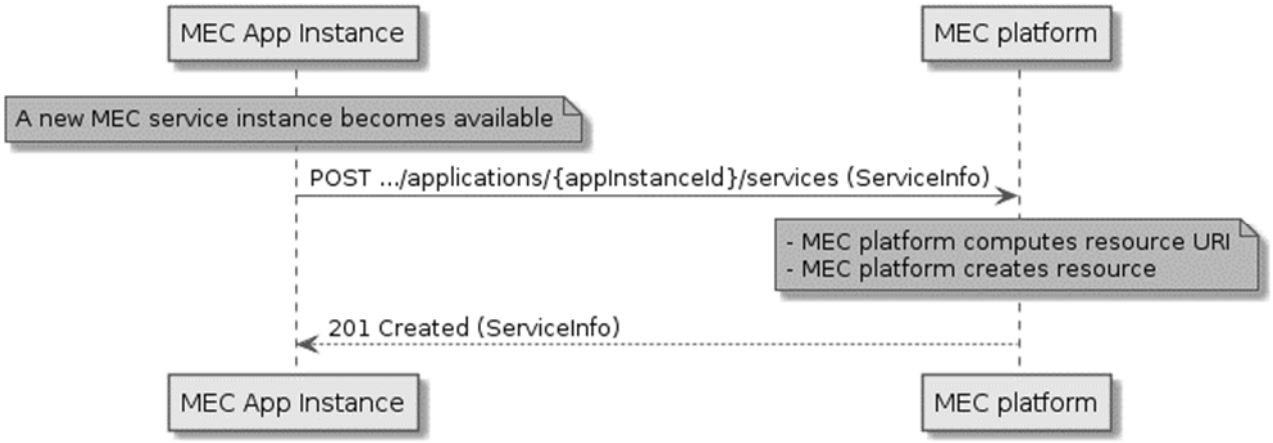


Figure 8.2.6.3.4-1: New service registration

POST HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.6.3.4-1 and 8.2.6.3.4-2.

Table 8.2.6.3.4-1: URI query parameters supported by the POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.6.3.4-2: Data structures supported by the POST request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	ServiceInfo	1	Message content in the request contains ServiceInfo to be created.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceInfo	1	201 Created	Upon success, the HTTP response shall include a "Location" HTTP header that contains the resource URI of the created resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.6.3.5 DELETE

Not supported.

8.2.7 Resource: individual mecService of an application instance

8.2.7.1 Description

This resource is used to represent a MEC service instance that is associated with an application instance, which follows the resource data type of "ServiceInfo" as specified in clause 8.1.2.2.

8.2.7.2 Resource definition

Resource URI: {apiRoot}/mec_service_mgmt/v1/applications/{appInstanceId}/services/{serviceId}

Resource URI variables for this resource are defined in table 8.2.7.2-1.

Table 8.2.7.2-1: Resource URI variables for resource "individual mecService of an application instance"

Name	Definition
apiRoot	See clause 8.2.2.
appInstanceId	Represents a MEC application instance.
serviceId	Represents a MEC service instance (see note).
NOTE: serviceId corresponds to serInstanceId (defined in table 8.1.2.2-1).	

8.2.7.3 Resource methods

8.2.7.3.1 GET

This method retrieves information about a mecService resource that is associated with an application instance. This method is typically used in "service availability query" procedure as described in clause 5.2.5.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.7.3.1-1 and 8.2.7.3.1-2.

Table 8.2.7.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.7.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceInfo	1	200 OK	It is used to indicate nonspecific success. The response message content contains a representation of the resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.7.3.2 PUT

This method updates the information about a mecService resource that is associated with the application instance. As specified in ETSI GS MEC 009 [5], the PUT HTTP method has "replace" semantics.

PUT method is typically used in "service availability update" procedure as described in clause 5.2.4. Figure 8.2.7.3.2-1 shows the message flow using PUT.

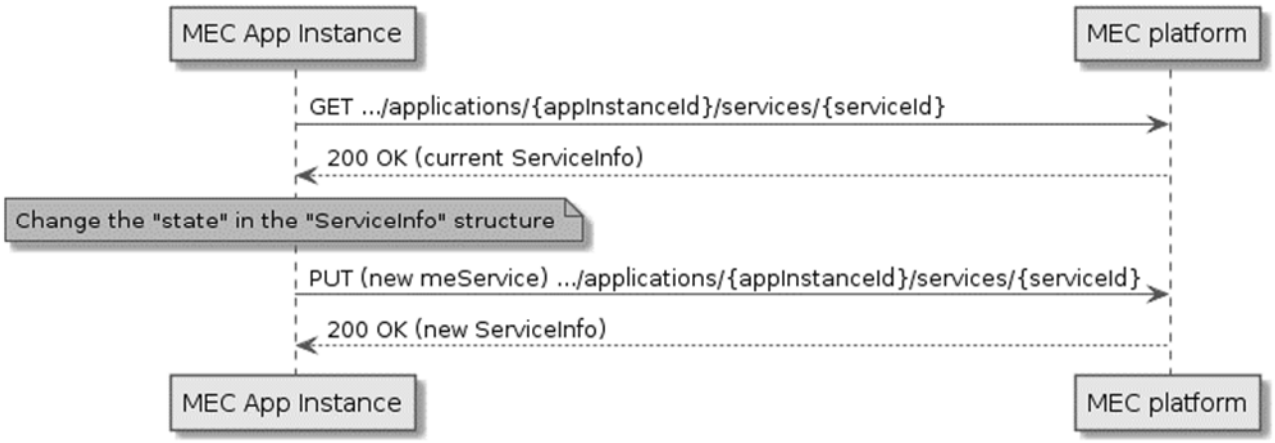


Figure 8.2.7.3.2-1: Service availability update using PUT

PUT HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.7.3.2-1 and 8.2.7.3.2-2.

Table 8.2.7.3.2-1: URI query parameters supported by the PUT method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.7.3.2-2: Data structures supported by the PUT request/response on this resource

	Data type	Cardinality	Response codes	Remarks
Request message content	ServiceInfo	1		One or more updated attributes that are allowed to be changed (i.e. "state" or other attributes based on definition in clause 8.1.2.2) are included in the ServiceInfo data structure in the message content of the request.
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceInfo	1	200 OK	Upon success, a response message content containing data type describing the updated ServiceInfo is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	412 Precondition Failed	It is used when a condition has failed during conditional requests, e.g. when using Etags to avoid write conflicts. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

8.2.7.3.3 PATCH

Not supported.

8.2.7.3.4 POST

Not supported.

8.2.7.3.5 DELETE

This method deletes a mecService resource. This method is typically used in the service deregistration procedure. Figure 8.2.7.3.5-1 shows the example message flows using DELETE method.

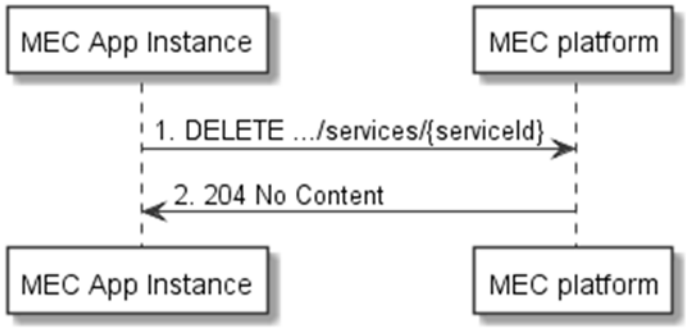


Figure 8.2.7.3.5-1: Service deregistration

DELETE HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.7.3.5-1 and 8.2.7.3.5-2.

Table 8.2.7.3.5-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.7.3.5-2: Data structures supported by the DELETE request on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	n/a		204 No Content	The operation has been successful. The response message content shall be empty.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.8 Resource: all mecSrvMgmtSubscription

8.2.8.1 Description

This resource is used to represent all subscriptions of a subscriber to the notifications from the MEC platform.

8.2.8.2 Resource definition

Resource URI: {apiRoot}/mec_service_mgmt/v1/applications/{appInstanceId}/subscriptions

Resource URI variables for this resource are defined in table 8.2.8.2-1.

Table 8.2.8.2-1: Resource URI variables for resource "all mecSrvMgmtSubscription"

Name	Definition
apiRoot	See clause 8.2.2.
appInstanceId	Represents a MEC application instance. Note that the appInstanceId is allocated by the MEC platform manager.

8.2.8.3 Resource methods

8.2.8.3.1 GET

The GET method may be used to request information about all subscriptions for this requestor. Upon success, the response contains message content with all the subscriptions for the requestor.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.8.3.1-1 and 8.2.8.3.1-2.

Table 8.2.8.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.8.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	SubscriptionLinkList	1	200 OK	Upon success, a response message content containing the list of links to the requested subscriptions is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.8.3.2 PUT

Not supported.

8.2.8.3.3 PATCH

Not supported.

8.2.8.3.4 POST

The POST method may be used to create a new subscription. One example use case is to create a new subscription to the MEC service availability notifications. Upon success, the response contains message content describing the created subscription. This method is typically used in "Subscribing to service availability event notifications" procedure as described in clause 5.2.6.2. Figure 8.2.8.3.4-1 shows the example message flows using POST method.

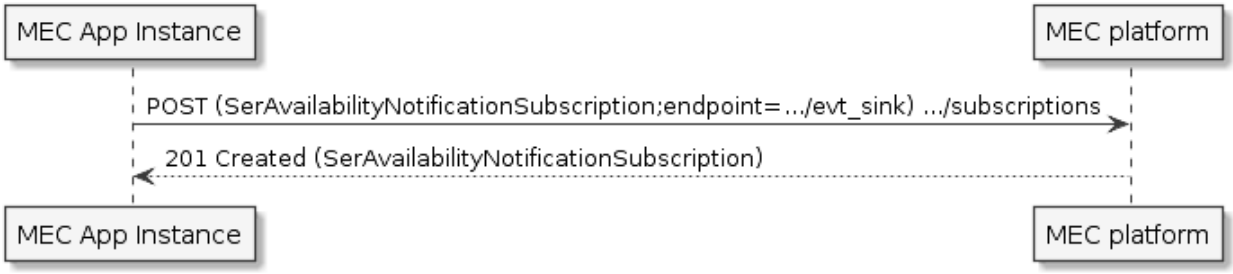


Figure 8.2.8.3.4-1: Subscribing to service availability event notifications

POST HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.8.3.4-1 and 8.2.8.3.4-2.

Table 8.2.8.3.4-1: URI query parameters supported by the POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.8.3.4-2: Data structures supported by the POST request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	SerAvailabilityNotificationSubscription	1	Message content in the request contains a subscription to the MEC service availability notifications that is to be created.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	SerAvailabilityNotificationSubscription	1	201 Created	Upon success, the HTTP response shall include a "Location" HTTP header that contains the resource URI of the created subscription resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.8.3.5 DELETE

Not supported.

8.2.9 Resource: individual mecSrvMgmtSubscription

8.2.9.1 Description

This resource is used to represent a subscription to the notifications from the MEC platform. When this resource represents a subscription to the notifications regarding the availability of a MEC service or a list of MEC services, it shall follow the data type of "SerAvailabilityNotificationSubscription" as specified in clause 8.1.3.2. The notifications that are related to a meSerAvailSubscription follow the data type of "ServiceAvailabilityNotification" as specified in clause 8.1.4.2.

8.2.9.2 Resource definition

Resource URI: {apiRoot}/mec_service_mgmt/v1/applications/{appInstanceId}/subscriptions/{subscriptionId}

Resource URI variables for this resource are defined in table 8.2.9.2-1.

Table 8.2.9.2-1: Resource URI variables for resource "individual mecSrvMgmtSubscription"

Name	Definition
apiRoot	See clause 8.2.2.
appInstanceId	Represents a MEC application instance. Note that the appInstanceId is allocated by the MEC platform manager.
subscriptionId	Represents a subscription to the notifications from the MEC platform.

8.2.9.3 Resource methods

8.2.9.3.1 GET

The GET method requests information about a subscription for this requestor. Upon success, the response contains message content with the subscription for the requestor.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.9.3.1-1 and 8.2.9.3.1-2.

Table 8.2.9.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.9.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	SerAvailabilityNotification Subscription	1	200 OK	Upon success, a response message content containing the requested subscription is returned.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.9.3.2 PUT

Not supported.

8.2.9.3.3 PATCH

Not supported.

8.2.9.3.4 POST

Not supported.

8.2.9.3.5 DELETE

This method deletes a mecSrvMgmtSubscription. This method is typically used in "Unsubscribing from event notifications" procedure as described in clause 5.2.6.3. Figure 8.2.9.3.5-1 shows the example message flows using DELETE method.

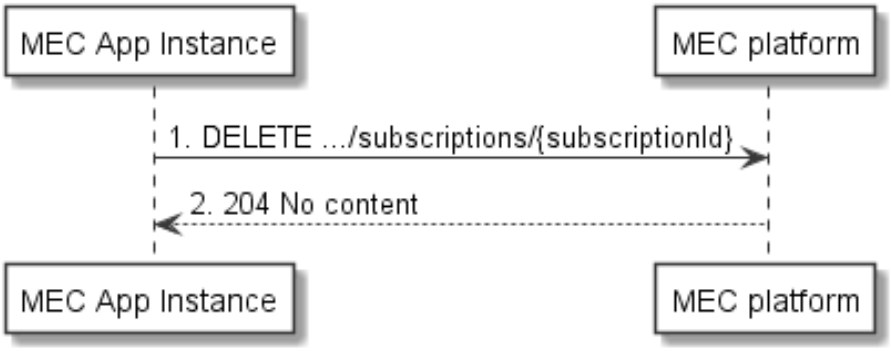


Figure 8.2.9.3.5-1: Unsubscribing from MEC service management event notifications

DELETE HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.9.3.5-1 and 8.2.9.3.5-2.

Table 8.2.9.3.5-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.9.3.5-2: Data structures supported by the DELETE request on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	n/a		204 No Content	
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.9a Resource: notification callback

8.2.9a.1 Description

This resource represents a notification callback, which can be used by the MEC Platform to send notifications related to service management events to a subscribed MEC application instance.

8.2.9a.2 Resource definition

The resource of callback URI is provided by the subscriber when subscribing to the notification.

8.2.9a.3 Resource methods

8.2.9a.3.1 GET

Not supported.

8.2.9a.3.2 PUT

Not supported.

8.2.9a.3.3 PATCH

Not supported.

8.2.9a.3.4 POST

The POST method delivers a notification from the MEC Platform to the subscriber.

This method shall follow the provisions specified in tables 8.2.9a.3.4-1 and 8.2.9a.3.4-2 for URI parameters, request and response data structures, and response codes.

Table 8.2.9a.3.4-1: URI query parameters supported by POST method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.9a.3.4-2: Data structures supported by POST request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	ServiceAvailabilityNotification	1	A notification for service availability update.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	n/a		204 No Content	The notification was delivered successfully. The response message content shall be empty.
	ProblemDetails	0..1	401 Unauthorized	It is used when the client did not submit credentials. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	429 Too Many Requests	It is used when a rate limiter has triggered. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

8.2.9a.3.5 DELETE

Not supported.

8.2.10 Resource: individual mecServiceLiveness

8.2.10.1 Description

This resource is used to represent the liveness of a MEC service instance produced by an application instance.

8.2.10.2 Resource definition

Resource URI: (allocated by the MEC platform)

The URI of this resource is allocated by the MEC platform at the time of MEC service registration and signalled in the "liveness" link in the representation of the related "Individual mecService of an application instance" resource.

Resource URI variables for this resource are defined in table 8.2.10.2-1.

Table 8.2.10.2-1: Resource URI variables for resource "individual mecServiceLiveness"

Name	Definition
none specified	

8.2.10.3 Resource methods

8.2.10.3.1 GET

This method retrieves information about an "Individual mecServiceLiveness" resource.

This method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.10.3.1-1 and 8.2.10.3.1-2.

Table 8.2.10.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.10.3.1-2: Data structures supported by the GET request/response on this resource

Request message content	Data type	Cardinality	Remarks	
	n/a			
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceLivenessInfo	1	200 OK	It is used to indicate nonspecific success. The response message content contains a representation of the resource.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.

8.2.10.3.2 PUT

Not supported.

8.2.10.3.3 PATCH

As specified in ETSI GS MEC 009 [5], the PATCH HTTP method (see IETF RFC 5789 [16]) is used to update a resource on top of the existing resource state with partial changes described by the client. As opposed to PUT, PATCH does not carry a representation of the resource in the message content, but a "deltas document" (see the definition of the "ServiceLivenessUpdate" type in clause 8.1.2.5) that instructs the server how to modify the resource representation.

The PATCH method is used in the "Service heartbeat" procedure as described in clause 5.2.12 and is referred to also as "heartbeat" message. Figure 8.2.10.3.3-1 shows the message flow using PATCH. It is the intent of this message to (re)confirm the "ACTIVE" state, but not to change the state from "INACTIVE" to "ACTIVE".

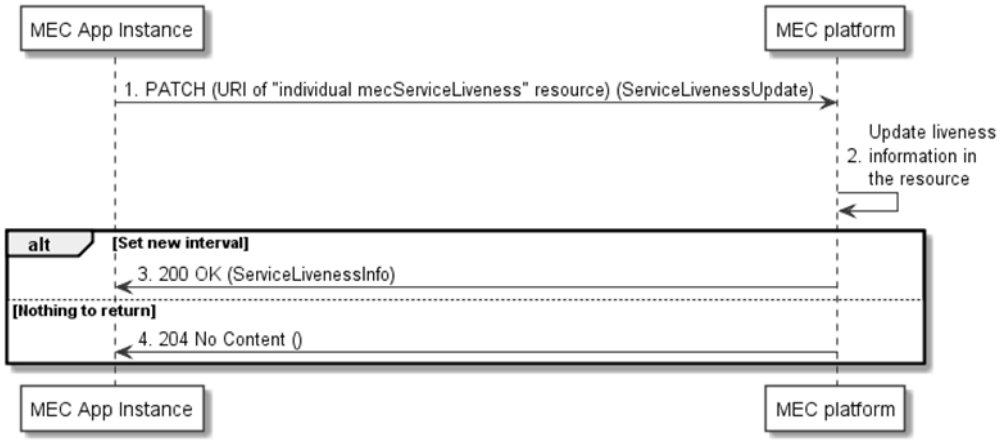


Figure 8.2.10.3.3-1: Service liveness update using PATCH

1) The MEC application instance that provides MEC service shall send a PATCH request to the resource URI representing the liveness of the service instance.

2) The MEC platform shall update the liveness resource as follows: It shall record the time when the message was received in the "timeStamp" attribute. Also, if the "state" attribute in the resource contains the value "SUSPENDED" and the "state" attribute in the message content contains the value "ACTIVE", it shall set the value of the "state" attribute in the resource to that value.

3) If there is no message content to return upon successful execution, the MEC platform shall return "204 No Content".

4) Alternatively, if the MEC platform intends to instruct the application to use a new liveness "interval" value for the service instance, it shall return "200 OK" along with the full ServiceLivenessInfo.

Error condition: Overwriting the "INACTIVE" state in a "heartbeat" message is forbidden and results in an error.

The PATCH HTTP method shall comply with the URI query parameters, request and response data structures, and response codes, as specified in tables 8.2.10.3.3-1 and 8.2.10.3.3-2.

Table 8.2.10.3.3-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	Cardinality	Remarks
n/a			

Table 8.2.10.3.3-2: Data structures supported by the PATCH request/response on this resource

Request message content	**Data type**	**Cardinality**	**Remarks**	
	ServiceLivenessUpdate	1	It contains an update of the liveness state.	
Response message content	**Data type**	**Cardinality**	**Response codes**	**Remarks**
	ServiceLivenessInfo	1	200 OK	Upon success, a response message content is returned containing the updated liveness interval value of the service Instance.
	n/a		204 No Content	Successful response sent when there is no need to provide a new liveness interval value to the service Instance.
	ProblemDetails	0..1	400 Bad Request	It is used to indicate that incorrect parameters were passed to the request. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	404 Not Found	It is used when a client provided a URI that cannot be mapped to a valid resource URI. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.
	ProblemDetails	0..1	403 Forbidden	The operation is not allowed given the current status of the resource. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	1	409 Conflict	The operation is not allowed due to a conflict with the state of the resource. The MEC platform shall respond with this code if the service instance is in "INACTIVE" state. More information shall be provided in the "detail" attribute of the "ProblemDetails" structure.
	ProblemDetails	0..1	412 Precondition Failed	It is used when a condition has failed during conditional requests, e.g. when using Etags to avoid write conflicts. In the returned ProblemDetails structure, the "detail" attribute should convey more information about the error.

8.2.10.3.4 POST

Not supported.

8.2.10.3.5 DELETE

Not supported.

9 MEC service management realized by CAPIF APIs

9.0 Introduction (informative)

3GPP has developed a common API framework (CAPIF) that includes common aspects applicable to any northbound service APIs. The common API framework applies to both 4G and 5G systems, and is independent of the underlying 3GPP access technologies (e.g. E-UTRA, NR).

ETSI TS 123 222 [23] specifies the architecture, procedures and information flows necessary for the CAPIF, while ETSI TS 129 222 [21] describes the CAPIF protocol. The CAPIF functional model is organized into functional entities to describe a functional architecture which enables an API invoker to access and invoke service APIs. The relationship between the MEC API framework and the CAPIF is shown in figure 9.0-1.

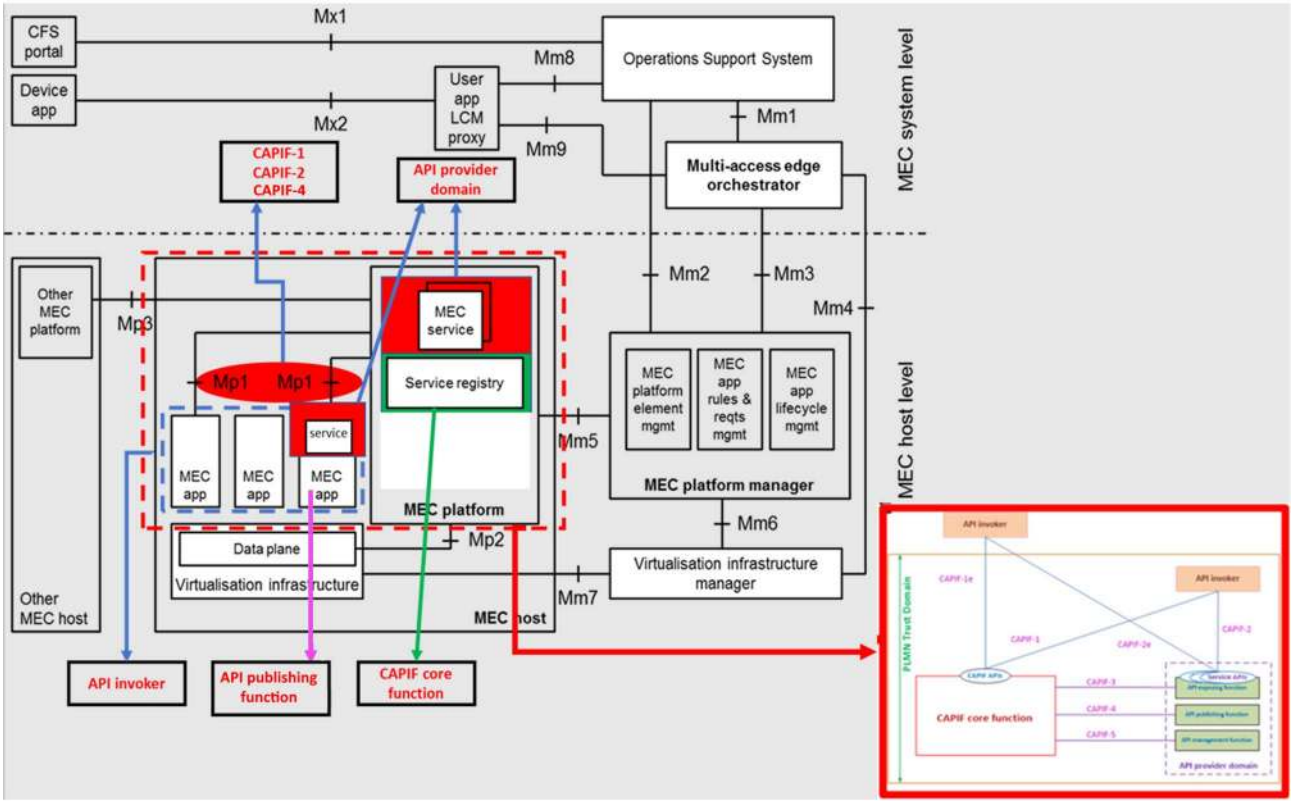


Figure 9.0-1: Relationship between MEC and 3GPP CAPIF

The MEC platform includes API related functionality such as service registry, which is equivalent to the API registry of the CAPIF core function. The existing MEC platform functionality related to API enablement can be mapped into the CAPIF core function.

The API provider domain in CAPIF collectively represents the service APIs available for consumption in any 5G network functions and any trusted 3rd party application functions. A MEC service produced by a MEC application or the MEC platform can be mapped into the API provider domain in CAPIF. A MEC application or MEC platform consuming a service corresponds to an API invoker in CAPIF.

MEC deployments can reuse CAPIF functionality to harmonize between MEC and CAPIF, e.g. to integrate MEC applications into a joint 3GPP-MEC deployment. To enable such harmonization, this clause defines a variant of the MEC service management functionality (see clause 8) that leverages the CAPIF Publish Service API, CAPIF Discover Service API and CAPIF Events API (see ETSI TS 129 222 [21]). This variant is herein referred to as the "MEC profile of CAPIF".

CAPIF inherently supports the signalling of REST interfaces, whilst the defined MEC profile of CAPIF adds support for alternative transports based on the CAPIF extension mechanisms. When using the MEC profile of CAPIF defined in the present document, the producers of the MEC services (MEC platform or MEC apps) take the role of API Exposing Functions (AEFs) whereas the consumers of the MEC services (again, MEC platform or MEC apps) take the role of API invokers. Further, MEC applications registering/publishing MEC services they produce take the role of API Publishing Functions (APFs).

Given that CAPIF is extensible, the CAPIF Publish Service API, CAPIF Discover Service API and CAPIF Events API can be consumed from a CAPIF Core Function (CCF) that supports the extensibility mechanisms as defined in ETSI TS 129 222 [21], i.e. which supports the CAPIF features ExtendedIntfDesc, MultipleCustomOperations, VendorExt and VendSpecQueryParams. Alternatively, the MEC Platform can produce these APIs as profiled in the present document (and termed as the "MEC profile of CAPIF").

9.1 Data model

9.1.1 Introduction

The data model from ETSI TS 129 222 [21] is reused.

The following resource and notification data types and all data types recursively referenced from them are reused, with the modifications defined in the subsequent clauses:

- DiscoveredAPIs
- ServiceAPIDescription
- EventSubscription
- EventNotification

NOTE: Certain CAPIF data types are reused without modifications to the data type itself, as defined in the subsequent clauses. However, some of these data types can recursively reference further reused CAPIF data types which are modified in the MEC context, as defined in clauses 9.1.2.2 and 9.1.5.

9.1.2 Resource data types

9.1.2.1 Type: DiscoveredAPIs

The DiscoveredAPIs data type shall be supported as defined in clause 8.1.4.2.2 of ETSI TS 129 222 [21]. There are no MEC related modifications to this particular data type.

9.1.2.2 Type: ServiceAPIDescription

The ServiceAPIDescription data type shall be supported as defined in clause 8.2.4.2.2 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.2.2-1.

Table 9.1.2.2-1: Profiling of the ServiceAPIDescription data type

Attribute name	Modifications
apiName	Shall be set to the value of the "serName" attribute as defined in clause 8.1.2.2.
apiId	If present, shall be set to the value of the "serInstanceId" attribute as defined in clause 8.1.2.2.
shareableInfo	See note.
serviceAPICategory	This attribute shall be set to the same value as the attribute "id" in "CategoryRef".
ccfId	See note.
apiSuppFeats	See note.
pubApiPath	See note.
apiStatus	See note.
vendorSpecific-urn:etsi:mec:capifext:service-info	This additional attribute of data type MecServiceInfoCapifExt (see clause 9.1.5.1) shall be included to carry the MEC specific extensions. This requires support for the CAPIF feature "VendorExt" as defined in clause 8.2.6 of ETSI TS 129 222 [21].
NOTE: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context.	

9.1.2.3 Type: ServiceAPIDescriptionPatch

The ServiceAPIDescriptionPatch data type shall be supported as defined in clause 8.2.4.2.11 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.2.3-1.

Table 9.1.2.3-1: Profiling of the ServiceAPIDescriptionPatch data type

Attribute name	Modifications
shareableInfo	See note.
serviceAPICategory	This attribute shall be set to the same value as the attribute "id" in "CategoryRef".
ccfId	See note.
apiSuppFeats	See note.
pubApiPath	See note.
apiStatus	See note.
vendorSpecific-urn:etsi:mec:capifext:service-info	This additional attribute of data type MecServiceInfoCapifExtPatch (see clause 9.1.5.8) shall be included to carry the MEC specific extension attributes to be changed. This requires support for the CAPIF feature "VendorExt" as defined in clause 8.2.6 of ETSI TS 129 222 [21].
NOTE: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context.	

9.1.3 Subscription data types

9.1.3.1 Type: EventSubscription

The EventSubscription data type shall be supported as defined in clause 8.3.4.2.2 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.3.1-1.

Table 9.1.3.1-1: Profiling of the EventSubscription data type

Attribute name	Modifications
events	The applicable subset of the values in the type "CAPIFEvent" and its mapping to the related MEC event types is defined in clause 9.1.6.1.
eventReq	See note.
notificationDestination	Shall be set to the value of the "callbackReference" attribute in the "SerAvailabilityNotificationSubscription" structure as defined in clause 8.1.3.2.
requestTestNotification	This attribute is aligned with the provisions in ETSI GS MEC 009 [5] which refer to the related mechanism from 3GPP.
websocketNotifConfig	This attribute is aligned with the provisions in ETSI GS MEC 009 [5] which refer to the related mechanism from 3GPP.
NOTE: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context.	

9.1.3.2 Type: EventSubscriptionPatch

The EventSubscriptionPatch data type shall be supported as defined in clause 8.3.4.2.8 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.3.2-1.

Table 9.1.3.2-1: Profiling of the EventSubscriptionPatch data type

Attribute name	Modifications
events	The applicable subset of the values in the type "CAPIFEvent" and its mapping to the related MEC event types is defined in clause 9.1.6.1.
eventReq	See note.
notificationDestination	Shall be set to the value of the "callbackReference" attribute in the "SerAvailabilityNotificationSubscription" structure as defined in clause 8.1.3.2.
NOTE: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context.	

9.1.4 Notification data types

9.1.4.1 Type: EventNotification

The EventNotification data type shall be supported as defined in clause 8.3.4.2.3 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.4.1-1.

Table 9.1.4.1-1: Profiling of the EventNotification data type

Attribute name	Modifications
events	The values SERVICE_API_AVAILABLE, SERVICE_API_UNAVAILABLE and SERVICE_API_UPDATE defined in the type "CAPIFEvent" (see clause 8.3.4.3.3 of ETSI TS 129 222 [21]) shall be supported. The remaining values of that type need not be supported.

9.1.5 Referenced structured data types

9.1.5.1 Type: MecServiceInfoCapifExt

This data type defines the MEC specific CAPIF extensions for the service API description based on the attributes of the ServiceInfo data type defined in clause 8.1.2.2, and data types referenced from it.

Table 9.1.5.1-1: Attributes of the MecServiceInfoCapifExt data type

Attribute name	Data type	Cardinality	Description
serializer	SerializerType	1	As defined in clause 8.1.2.2.
state	ServiceState	1	As defined in clause 8.1.2.2.
scopeOfLocality	LocalityType	0..1	As defined in clause 8.1.2.2.
consumedLocalOnly	Boolean	0..1	As defined in clause 8.1.2.2.
isLocal	Boolean	0..1	As defined in clause 8.1.2.2.
category	CategoryRef	1	As defined in clause 8.1.5.2.

9.1.5.2 Type: MecTransportInfoCapifExt

This data type defines the MEC specific CAPIF extensions related to alternative transports based on the attributes of the TransportInfo data type defined in clause 8.1.2.3.

Table 9.1.5.2-1: Attributes of the MecTransportInfoCapifExt

Attribute name	Data type	Cardinality	Description
name	String	1	As defined in clause 8.1.2.3.
description	String	0..1	As defined in clause 8.1.2.3.
type	TransportType	1	As defined in clause 8.1.2.3.
protocol	String	1	As defined in clause 8.1.2.3.
version	String	1	As defined in clause 8.1.2.3.
security	SecurityInfo	1	As defined in clause 8.1.2.3.
implSpecificInfo	Not specified	0..1	As defined in clause 8.1.2.3.

9.1.5.3 Type: AefProfile

The AefProfile data type shall be supported as defined in clause 8.2.4.2.4 of ETSI TS 129 222 [21] with the MEC-related modifications specified in table 9.1.5.3-1.

Table 9.1.5.3-1: Profiling of the AefProfile data type

Attribute name	Modifications
aefId	Shall be set to the value of the "id" attribute as defined in clause 8.1.2.3.
versions	This array shall contain a single entry.
protocol	See note 1.
dataFormat	See note 1.
securityMethods	See note 1.
domainName	This attribute shall not be supported by an ETSI MEC implementation. See note 2.
interfaceDescriptions	This attribute shall be supported by an ETSI MEC implementation. See note 2.
aefLocation	This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context.
vendorSpecific-urn:etsi:mec:capifext:transport-info	This additional attribute of data type MecTransportInfoCapifExt (see clause 9.1.5.2) shall be included. This requires support for the CAPIF feature "VendorExt" as defined in clause 8.2.6 of ETSI TS 129 222 [21].
NOTE 1: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context. See also clause 9.3 on the handling of this attribute in joint 3GPP-MEC deployments. NOTE 2: The "interfaceDescriptions" attribute provides a more general alternative to the "domainName" attribute.	

9.1.5.4 Type: Version

The Version data type shall be supported as defined in clause 8.2.4.2.5 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.5.4-1.

Table 9.1.5.4-1: Profiling of the Version data type

Attribute name	Modifications
apiVersion	As defined for the "version" attribute in clause 8.1.2.2.

9.1.5.5 Type: InterfaceDescription

The InterfaceDescription data type shall be supported as defined in clause 8.2.4.2.3 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.5.5-1.

Table 9.1.5.5-1: Profiling of the InterfaceDescription data type

Attribute name	Modifications
securityMethods	This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context. See also clause 9.3 on the handling of this attribute in joint 3GPP-MEC deployments.

9.1.5.6 Type: CAPIFEventDetail

The CAPIFEventDetail data type shall be supported as defined in clause 8.3.4.2.5 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.5.6-1.

Table 9.1.5.6-1: Profiling of the CAPIFEventDetail data type

Attribute name	Modifications
apiIds	As defined for the "serInstanceIds" attribute in clause 8.1.3.2.
apiInvokerIds	See note.
accCtrlPolList	See note.
invocationLogs	See note.
apiTopoHide	See note.
NOTE: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context.	

9.1.5.7 Type: CAPIFEventFilter

The CAPIFEventFilter data type shall be supported as defined in clause 8.3.4.2.4 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.5.7-1.

Table 9.1.5.7-1: Profiling of the CAPIFEventFilter data type

Attribute name	Modifications
apiIds	As defined for the "serInstanceIds" attribute in clause 8.1.3.2.
apiInvokerIds	See note.
aefIds	See note.
NOTE: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context.	

9.1.5.8 Type: MecServiceInfoCapifExtPatch

This data type defines changes when patching the MEC specific CAPIF extensions for the service API description based on the attributes of the ServiceInfo data type defined in clause 8.1.2.2, and data types referenced from it.

Table 9.1.5.8-1: Attributes of the MecServiceApiDescCapifExtPatch data type

Attribute name	Data type	Cardinality	Description
serializer	SerializerType	0..1	As defined in clause 8.1.2.2.
state	ServiceState	0..1	As defined in clause 8.1.2.2.
scopeOfLocality	LocalityType	0..1	As defined in clause 8.1.2.2.
consumedLocalOnly	Boolean	0..1	As defined in clause 8.1.2.2.
isLocal	Boolean	0..1	As defined in clause 8.1.2.2.
category	CategoryRef	0..1	As defined in clause 8.1.5.2.

9.1.6 Referenced simple data types and enumerations

9.1.6.1 Enumeration: CAPIFEvent

The CAPIFEvent data type shall be supported as defined in clause 8.3.4.3.3 of ETSI TS 129 222 [21] with the MEC related modifications specified in table 9.1.6.1-1.

Table 9.1.6.1-1: Profiling of the CAPIFEvent data type

Attribute name	Modifications
SERVICE_API_AVAILABLE	This value shall be supported. It maps to the value "ADDED" in clause 8.1.6.7.
SERVICE_API_UNAVAILABLE	This value shall be supported. It maps to the value "REMOVED" in clause 8.1.6.7.
SERVICE_API_UPDATE	This value shall be supported. It maps to the values "STATE_CHANGED" and "ATTRIBUTES_CHANGED" in clause 8.1.6.7.
(all other values)	These values need not be supported by an ETSI MEC implementation as they are not relevant for the part of the CAPIF functionality that is re-used by ETSI MEC.

9.2 API definition

9.2.1 Introduction

This clause defines how to reuse the resources defined in ETSI TS 129 222 [21] for the CAPIF Publish Service API, the CAPIF Discover Service API and the CAPIF Events API to realize the MEC service management functionality.

When producing MEC services, the MEC platform and the MEC app take the role of a CAPIF API Exposing Function (AEF). When registering services, the MEC platform and the MEC app take the role of a CAPIF API Publishing Function (APF). When discovering or consuming services, the MEC platform and the MEC app take the role of a CAPIF API invoker.

9.2.2 Global definitions and resource structure

The resource URIs are reused from CAPIF. As three APIs are reused, there are three different roots for the resource URIs:

{apiRoot}/service-apis/{apiVersion}/

{apiRoot}/published-apis/{apiVersion}/

{apiRoot}/capif-events/{apiVersion}/

The "apiRoot" is discovered by means outside the scope of the using the service registry. It includes the scheme ("https"), host and optional port, and an optional prefix string. The "apiVersion" is set to "v1" as defined in ETSI TS 129 222 [21].

When these APIs are produced by the MEC platform, the following applies:

1) The APIs shall support and use HTTP over TLS as defined in clause 6.22 of ETSI GS MEC 009 [5].

2) The APIs shall support and use Oauth 2.0, as defined in clause 6.16 of ETSI GS MEC 009 [5]. This Oauth 2.0 authorization procedure shall occur only on TLS-protected connections.

Figure 9.2.2-1 illustrates the resource URI structure of these APIs, as imported from ETSI TS 129 222 [21].

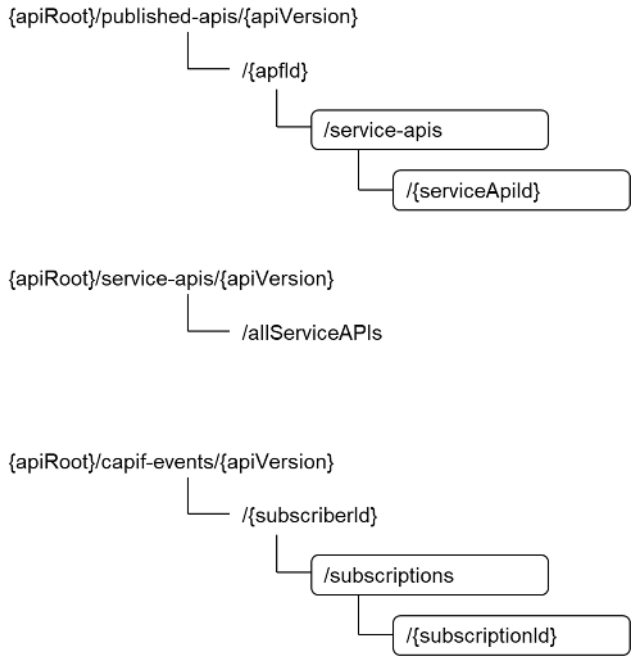


Figure 9.2.2-1: Resource URI structure of the MEC service management based on CAPIF

Table 9.2.2-1 provides illustrates the resources reused from CAPIF (see ETSI TS 129 222 [21]) for MEC service management, and the applicable HTTP methods.

Table 9.2.2-1: Resources and methods overview

Resource name	Resource URI	HTTP method	Meaning
All published service APIs	/service-apis/{apiVersion}/allServiceAPIs	GET	Discover published service APIs and retrieve a collection of APIs according to certain filter criteria.
APF published APIs	/published-apis/{apiVersion}/{apfId}/service-apis	POST	Publish a new API.
		GET	Retrieve all published service APIs.
Individual APF published API	/published-apis/{apiVersion}/{apfId}/service-apis/{serviceApiId}	GET	Retrieve a published service API.
		PUT	Update a published service API.
		PATCH	Modify a published service API.
		DELETE	Unpublish a published service API.
CAPIF Events Subscriptions	/capif-events/{apiVersion}/{subscriberId}/subscriptions	POST	Creates a new individual CAPIF Event Subscription.
Individual CAPIF Events Subscription	/capif-events/{apiVersion}/{subscriberId}/subscriptions/{subscriptionId}	DELETE	Deletes an individual CAPIF Event Subscription identified by the subscriptionId.
		PUT	Updates an individual CAPIF Event Subscription identified by the subscriptionId, using replace semantics.

Resource name	Resource URI	HTTP method	Meaning
		PATCH	Updates an individual CAPIF Event Subscription identified by the subscriptionId, using modify semantics.
Notification callback	Client provided callback reference	POST	Send a notification.

9.2.3 Resource: All published service APIs

9.2.3.1 Description

This resource allows to discover service APIs.

9.2.3.2 Resource definition

The resource shall be supported as specified in clause 8.1.2.2 of ETSI TS 129 222 [21].

The resource URI is: {apiRoot}/service-apis/{apiVersion}/allServiceAPIs

9.2.3.3 Resource methods

9.2.3.3.1 GET

This method shall be supported as specified in clause 8.1.2.2.3.1 of ETSI TS 129 222 [21], with the profiling of the resource URI query parameters for use by MEC as defined in table 9.2.3.3.1-1.

Table 9.2.3.3.1-1: Profiling of the URI query parameters

URI query parameter	Modifications
api-invoker-id	If the Discover_Service_API is produced by the MEC platform towards the MEC applications, this identifier shall be set to the value of the "appInstanceId" (see clause 7.1.2.6). If the Discover_Service_API is produced by the CCF, the provisions in clause 8.1.2.2.3.1 of ETSI TS 129 222 [21] apply.
api-name	This query parameter corresponds to the parameter "ser_name" as defined in clause 8.2.3.3.1. As opposed to the cardinality of ser_instance_id defined in clause 8.2.3.3.1 that is 0..N, this parameter only supports a cardinality of 0..1. See note 3.
api-version	See note 1.
comm-type	See note 1.
protocol	See note 1.
aef-id	See note 1.
data-format	See note 1.
api-cat	Same as vend-spec-etsi-mec-ser-category-id.
preferred-aef-loc	See note 1.
api-supported-features	See note 1.
vend-spec-etsi-mec-ser-instance-id	This vendor specific API discovery query filter parameter corresponds to the query parameter "ser_instance_id" as defined in clause 8.2.3.3.1. The "target" attribute shall be set to "/apiId". As opposed to the cardinality of ser_instance_id defined in clause 8.2.3.3.1 that is 0..N, this parameter only supports a cardinality of 0..1. See notes 2 and 3.
vend-spec-etsi-mec-ser-category-id	This vendor specific API discovery query filter parameter corresponds to the query parameter "ser_category_id" as defined in clause 8.2.3.3.1. The "target" attribute shall be set to "/vendorSpecific-urn:etsi:mec:capifext:service-info/serCategory/id". See notes 2, 3 and 4.
vend-spec-etsi-mec-scope-of-locality	This vendor specific API discovery query filter parameter corresponds to the query parameter "scope_of_locality" as defined in clause 8.2.3.3.1. The "target" attribute shall be set to "/vendorSpecific-urn:etsi:mec:capifext:service-info/scopeOfLocality". See notes 2 and 5.

URI query parameter	Modifications
vend-spec-etsi-mec-consumed-local-only	This vendor specific API discovery query filter parameter corresponds to the query parameter "consumed_local_only" as defined in clause 8.2.3.3.1. The "target" attribute shall be set to "/vendorSpecific-urn:etsi:mec: capifext:service-info/consumedLocalOnly". See notes 2 and 5.
vend-spec-etsi-mec-is-local	This vendor specific API discovery query filter parameter corresponds to the query parameter "consumed_local_only" as defined in clause 8.2.3.3.1. The "target" attribute shall be set to "/vendorSpecific-urn:etsi:mec: capifext:service-info/isLocal". See notes 2 and 5.
NOTE 1: This attribute need not be supported by an ETSI MEC implementation as it is not relevant in the MEC context. NOTE 2: The general mechanism of vendor specific API discovery query filter parameters is defined in note 2 in table 8.1.2.2.3.1-1 in ETSI TS 129 222 [21]. NOTE 3: In line with the definitions in clause 8.2.3.3.1, either "vend-spec-etsi-mec-ser-instance-id" or "api-name" or "vend-spec-etsi-mec-ser-category-id" or none of them shall be present. NOTE 4: This vendor specific API discovery query filter parameter shall be supported if the feature "VendSpecQueryParams" (see ETSI TS 129 222 [21] clause 8.1.6) is supported. NOTE 5: This vendor specific API discovery query filter parameter shall be supported if the features "VendSpecQueryParams" (see ETSI TS 129 222 [21] clause 8.1.6) and "VendorExt" (see ETSI TS 129 222 [21] clause 8.2.6) are supported.	

9.2.3.3.2 PUT

Not supported.

9.2.3.3.3 PATCH

Not supported.

9.2.3.3.4 POST

Not supported.

9.2.3.3.5 DELETE

Not supported.

9.2.4 Resource: APF published APIs

9.2.4.1 Description

This resource allows to discover publish (register) service APIs and to list the registered service APIs.

9.2.4.2 Resource definition

The resource shall be supported as specified in clause 8.2.2.2 of ETSI TS 129 222 [21], with the profiling of the URI variables for use by MEC as defined in table 9.2.4.2-1.

The resource URI is: {apiRoot}/published-apis/{apiVersion}/{apfId}/service-apis

Table 9.2.4.2-1: Profiling of the URI variables

URI variable name	Modifications
apfId	Identifier of the entity that registers the service API. If the MEC app plays the role of the APF, this variable shall be set to the value of the "appInstanceId" (see clause 7.1.2.6). If the MEC platform plays the role of the APF, this variable shall be set to a specific identifier that identifies the MEC platform.

9.2.4.3 Resource methods

9.2.4.3.1 GET

This method shall be supported as specified in clause 8.2.2.2.3.2 of ETSI TS 129 222 [21].

9.2.4.3.2 PUT

Not supported.

9.2.4.3.3 PATCH

Not supported.

9.2.4.3.4 POST

This method shall be supported as specified in clause 8.2.2.2.3.1 of ETSI TS 129 222 [21].

9.2.4.3.5 DELETE

Not supported.

9.2.5 Resource: Individual APF published API

9.2.5.1 Description

This resource allows to read, update and delete an individual service API publication (registration).

9.2.5.2 Resource definition

This resource shall be supported as specified in clause 8.2.2.3 of ETSI TS 129 222 [21], with the profiling of the URI variables for use by MEC as defined in table 9.2.5.2-1.

The resource URI is: {apiRoot}/published-apis/{apiVersion}/{apfId}/service-apis/{serviceApiId}

Table 9.2.5.2-1: Profiling of the URI variables

URI variable name	Modifications
apfId	Identifier of the entity that registers the service API. If the MEC app plays the role of the APF, this variable shall be set to the value of the "appInstanceId" (see clause 7.1.2.6). If the MEC platform plays the role of the APF, this variable shall be set to a specific identifier that identifies the MEC platform.
serviceApiId	No modifications.

9.2.5.3 Resource methods

9.2.5.3.1 GET

This method shall be supported as specified in clause 8.2.2.3.3.1 of ETSI TS 129 222 [21].

9.2.5.3.2 PUT

This method shall be supported as specified in clause 8.2.2.3.3.2 of ETSI TS 129 222 [21].

9.2.5.3.3 PATCH

This method shall be supported as specified in clause 8.2.2.3.3.4 of ETSI TS 129 222 [21].

9.2.5.3.4 POST

Not supported.

9.2.5.3.5 DELETE

This method shall be supported as specified in clause 8.2.2.3.3.3 of ETSI TS 129 222 [21].

9.2.6 Resource: CAPIF Events Subscriptions

9.2.6.1 Description

This resource allows to subscribe to notifications about CAPIF events. In this context, notifications are related to the publication (registration) of new services and to updates of existing service publications (registrations).

9.2.6.2 Resource definition

The resource shall be supported as specified in clause 8.3.2.2 of ETSI TS 129 222 [21], with the profiling of the URI variables for use by MEC as defined in table 9.2.6.2-1.

The resource URI is: {apiRoot}/capif-events/{apiVersion}/{subscriberId}/subscriptions

Table 9.2.6.2-1: Profiling of the URI variables

URI variable name	Modifications
subscriberId	This variable shall be set to the value of the "appInstanceId" (see clause 7.1.2.6).

9.2.6.3 Resource methods

9.2.6.3.1 GET

Not supported.

9.2.6.3.2 PUT

Not supported.

9.2.6.3.3 PATCH

Not supported.

9.2.6.3.4 POST

This method shall be supported as specified in clause 8.3.2.2.3.1 of ETSI TS 129 222 [21].

9.2.6.3.5 DELETE

Not supported.

9.2.7 Resource: Individual CAPIF Events Subscription

9.2.7.1 Description

This resource allows to delete and update a CAPIF event subscription.

9.2.7.2 Resource definition

The resource shall be supported as defined in clause 8.1.2.2.2 of ETSI TS 129 222 [21], with the profiling of the URI variables for use by MEC as defined in table 9.2.7.2-1.

The resource URI is:
{apiRoot}/service-apis/{apiVersion}/capif-events/{apiVersion}/{subscriberId}/subscriptions/{subscriptionId}

Table 9.2.7.2-1: Profiling of the URI variables

URI variable name	Modifications
subscriberId	This variable shall be set to the value of the "appInstanceId" (see clause 7.1.2.6).
subscriptionId	No modifications.

9.2.7.3 Resource methods

9.2.7.3.1 GET

Not supported.

9.2.7.3.2 PUT

This method shall be supported as specified in clause 8.3.2.3.3.2 of ETSI TS 129 222 [21].

9.2.7.3.3 PATCH

This method shall be supported as specified in clause 8.3.2.3.3.3 of ETSI TS 129 222 [21].

9.2.7.3.4 POST

Not supported.

9.2.7.3.5 DELETE

This method shall be supported as specified in clause 8.3.2.3.3.1 of ETSI TS 129 222 [21].

9.2.8 Resource: Notification callback

9.2.8.1 Description

This resource represents a notification callback the was provided by the MEC application upon subscription, which can be used by the MEC Platform to send notifications to a subscribed MEC application instance.

The delivery of notifications shall conform to clause 7.6 in ETSI TS 129 222 [21].

9.2.8.2 Resource definition

The resource of callback URI is provided by the subscriber when subscribing to the notification. The resource URI is determined by the subscriber.

9.2.8.3 Resource methods

9.2.8.3.1 GET

Not supported.

9.2.8.3.2 PUT

Not supported.

9.2.8.3.3 PATCH

Not supported.

9.2.8.3.4 POST

This method shall be supported as specified in clause 8.3.3 of ETSI TS 129 222 [21].

9.2.8.3.5 DELETE

Not supported.

9.3 CAPIF and MEC

9.3.1 Mapping of the resource structures (informative)

Table 9.3.1-1 shows the mapping of MEC service management API resources defined in the present document to CAPIF resources defined in ETSI TS 129 222 [21].

Table 9.3.1-1: Mapping of MEC resources and CAPIF resources for service management

MEC service management API		**CAPIF APIs [21]**	
resource name	**resource URI**	**resource name**	**resource URI**
Retrieve information about a list of mecService resources			
A list of mecService	mec_service_mgmt/v1/service	*CAPIF_Discover_Service_API:* All published service APIs	/service-apis/v1/allServiceApis
Retrieve information about a mecService resource			
Individual mecService	mec_service_mgmt/v1/services/{serviceId}	- (see note)	- (see note)
Retrieve information about the available transports			
A list of mecTransport	mec_service_mgmt/v1/transports	-	-
Retrieve information about a list of mecService resources of an application instance; Create a new individual mecService resource of an application instance ("register / publish service API")			
A list of mecService of an application instance	mec_service_mgmt/v1/applications/{appInstanceId}/services	CAPIF_Publish_Service_API: APF published APIs	/published-apis/v1/{apfId}/service-apis
Retrieve information about an individual mecService resource of an application instance; Update information about an individual mecService resource of an application instance; Delete information about an individual mecService resource of an application instance			
Individual mecService of an application instance	mec_service_mgmt/v1/applications/{appInstanceId}/services/{serviceId}	*CAPIF_Publish_Service_API:* Individual APF published API	/published-apis/v1/{apfId}/service-apis/{serviceApiId}
Retrieve information about a list of mecSrvMgmtSubscription resources for this subscriber			
Parent resource of all mecSrvMgmtSubscription of a subscriber	mec_service_mgmt/v1/applications/{appInstanceId}/subscriptions	*CAPIF_Events_API:* CAPIF Events Subscriptions	/capif-events/v1/{subscriberId}/subscriptions/
Retrieve information about an individual mecSrvMgmtSubscription resource for this subscriber			
Individual mecSrvMgmtSubscription	mec_service_mgmt/v1/applications/{appInstanceId}/subscriptions/{subscriptionId}	*CAPIF_Events_API:* Individual CAPIF Events Subscription	/capif-events/v1/{subscriberId}/subscriptions/{subscriptionId}
NOTE: Although there is no resource defined in CAPIF registry to represent individual services, the query of a specific service is possible by using suitable filtering parameters (see clause 9.2.3.3.1) with the CAPIF APIs.			

9.3.2 Mapping of the data models (informative)

9.3.2.1 Service data model

Figure 9.3.2.1-1 provides a UML class diagram which represents the CAPIF service data model with the MEC mappings specified in clause 9.1. CAPIF attributes that are needed for the system to work are depicted in black colour. If they have a correspondence to a MEC attribute, that MEC attribute is depicted in red next to the related CAPIF attribute. Classes and attributes that were added by MEC as CAPIF extensions are depicted with a name in red colour. CAPIF defined classes and attributes that contain information which has no correspondence in the MEC data model are depicted with a class name or attribute name in blue colour. These are typically being ignored by MEC implementations. The following CAPIF defined attributes for which MEC uses an alternative representation are depicted in blue colour and are stricken through: "protocol", "dataFormat", "apiStatus" and "securityMethods". CAPIF attributes related to specific CAPIF use cases which are not relevant for the use of CAPIF by MEC are depicted in black colour and are stricken through. These are also explicitly defined as not being part of the MEC profile of CAPIF in clause 9.1 (either "need not be supported" or "shall not be supported"). To express the constraint that only one of multiple attributes needs to be chosen when creating an instance of the class, the representation of a choice has been added to the standard UML representation.

Conceptually, the MEC "ServiceInfo" type maps to the CAPIF "ServiceAPIDescription" type and the MEC "TransportInfo" type maps to the CAPIF "AefProfile" type. The information represented in a single MEC "EndPointInfo" structure can be mapped to multiple structures of the CAPIF type "InterfaceDescription", with the difference that CAPIF does not support the use of an "alternative" format for endpoints. As "alternative" is anyway a place holder, this will be left for future specification if the need occurs. The MEC profile of CAPIF is more restrictive than CAPIF with regard to the mapping between AefProfile and Version: While CAPIF allows one or more Version entries mapped to a single AefProfile, the cardinality is restricted to 1 in the MEC profile of CAPIF.

CAPIF only maintains service API state when the ApiStatusMonitoring feature is supported. Otherwise, the approach in CAPIF is to delete a service API from the registry when the API becomes unavailable and republish it when it becomes available again. Such approach can lead to the API reappearing in the registry under another identifier than before which might confuse MEC clients. To represent status changes, the entity that has information about the API status (e.g. the API-producing application itself) updates the "state" attribute in the "MecServiceInfoCapifExt" structure in the CAPIF registry in case the MEC service API changes its state.

The CAPIF service registry does not support the negotiation of service liveness (i.e. heartbeat) checking which is defined in ETSI MEC by the ServiceInfo attributes "livenessInterval" and "_links/liveness". Such functionality is therefore not available in the present version of the MEC profile of CAPIF.

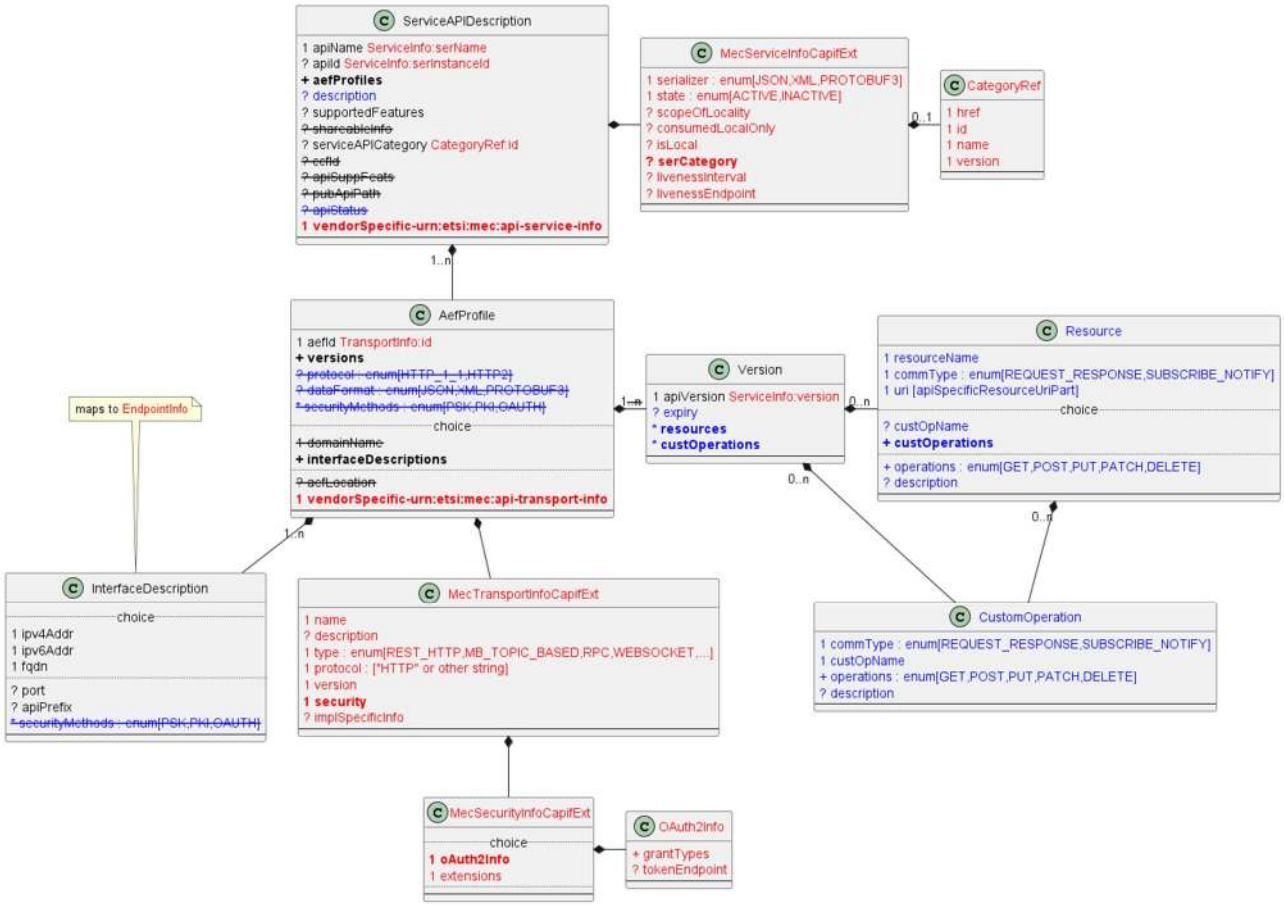


Figure 9.3.2.1-1: Mapping of the MEC service data model to the CAPIF service data model

9.3.2.2 Subscribe / notify

Figure 9.3.2.2-1 provides a UML class diagram which represents the CAPIF subscribe/notify model with the MEC mappings specified in clause 9.1.

There are a few differences:

1) CAPIF event filtering is coarser than MEC event filtering. For Service API related events, the only filter that CAPIF allows is apiId (mapped to serInstanceIds in MEC). All other MEC filters are not supported and the related filtering needs to be performed at API invoker side. Table 9.3.2.2-1 summarizes the CAPIF support of the MEC event filters.

Table 9.3.2.2-1: CAPIF support of the MEC event filters as defined in clause 8.1.3.2

MEC filter	CAPIF support	CAPIF attribute
serInstanceIds	supported	apiIds
serNames	not supported	-
serCategories	not supported	-
states	not supported	-
isLocal	not supported	-

2) CAPIF does not maintain service API state unless the feature ApiStatusMonitoring is supported. API state is represented as an additional extension attribute in the MEC profile of CAPIF, and an API state change will trigger a "SERVICE_API_UPDATE" CAPIF event like any other attribute change in the service API description. In contrast, MEC has two different event types for changing API state ("STATE_CHANGED") and changing any other API service API description attribute ("ATTRIBUTES_CHANGED"). In the MEC profile of CAPIF, both these MEC event types map to the same CAPIF event type "SERVICE_API_UPDATE".

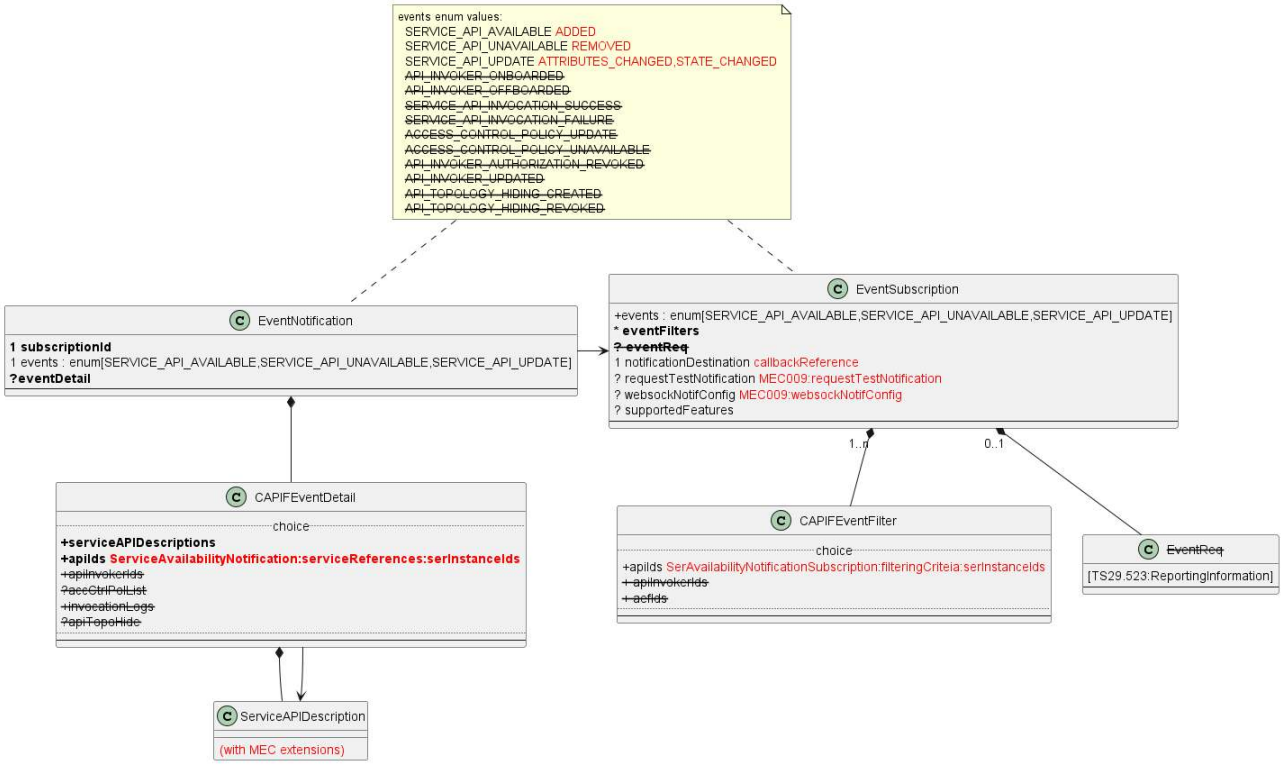


Figure 9.3.2.2-1: Mapping of the MEC subscribe/notify data model to the CAPIF subscribe/notify data model

9.3.2.3 Discovery queries

API discovery queries are realized in ETSI MEC and in 3GPP CAPIF by URI query parameters of the GET request that represents the set of registered service APIs. The mapping is not 1:1 however, as illustrated in figure 9.3.2.3-1.

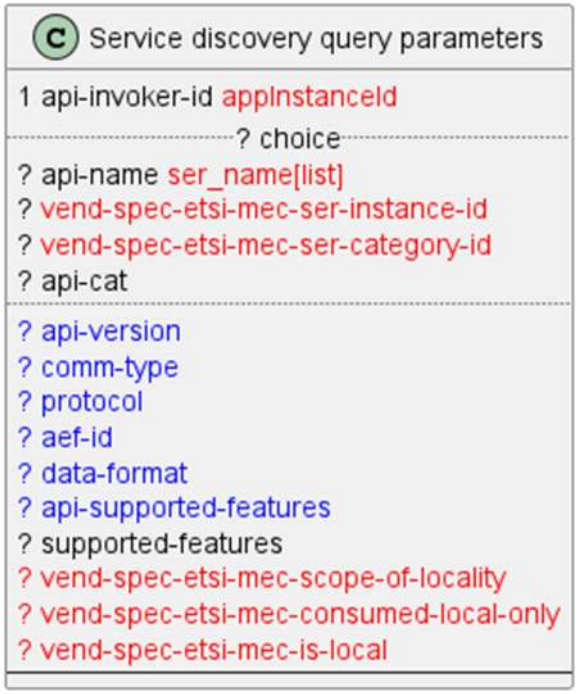


Figure 9.3.2.3-1: Mapping of the MEC and CAPIF service discovery query parameters

Annex A (informative): Complementary material for API utilization

To complement the definitions for each method and resource defined in the interface clauses of the present document, ETSI ISG MEC is providing for each MEC Platform Application Enablement API a supplementary description file compliant to the OpenAPI Specification [i.6].

In case of discrepancies between each supplementary description file and the related data structure definitions in the present document, the data structure definitions take precedence.

The supplementary description files, relating to the present document, are located at https://forge.etsi.org/rep/mec/gs011-app-enablement-api.

Annex B (informative): Void

Annex C (informative): Analysis of EASProfile

Table C-1 shows the analysis of the EASProfile attributes, which is specified in table 8.1.5.2.3-1 of ETSI TS 129 558 [i.11] (Release 17).

Table C-1: Analysis of the EASProfile attributes

Attribute name	Data type	P	Cardinality	Description	Remarks
easId	string	M	1	The application identifier of the EAS, e.g. URI, FQDN.	No existing MEC attribute defined. See note.
endPt	EndPoint	M	1	Endpoint information (URI, FQDN, IP address) used to communicate with the EAS. This information maybe discovered by EEC and exposed to Acs so that Acs can establish contact with the EAS.	Correspondingly, the attribute endpoint is defined in the Data Type *AppInfo* in the present document.
acIds	array(string)	O	1..N	Identities of the Application Clients that can be served by the EAS.	No existing MEC attribute defined. See note.
provId	string	O	0..1	Identifier of the ASP that provides the EAS.	Correspondingly, the attribute appProvider is defined in the Data Type *AppInfo* in the present document.
type	string	O	0..1	The category or type of EAS.	Similar to *CategoryRef* defined in the present document. See note.
scheds	array(SchedulEdCommunicationTime)	O	1..N	The availability schedule of the EAS.	Correspondingly, the attribute scheds is defined in the Data Type *AppInfo* in the present document.
svcArea	ServiceArea	O	0..1	The list of geographical and topological areas that the EAS serves. Acs in the UE that are outside the area will not be served.	Correspondingly, the attribute svcArea is defined in the Data Type *AppInfo* in the present document.
svcKpi	EASServiceKPI	O	0..1	Service characteristics provided by the EAS.	Correspondingly, the attribute svcKpi is defined in the Data Type *AppInfo* in the present document.
permLvl	array(string)	O	1..N	Level of service permissions supported by the EAS.	Correspondingly, the attribute permLvl is defined in the Data Type *AppInfo* in the present document.
easFeats	array(string)	O	1..N	Service specific features supported by the EAS (e.g. single vs multi-player gaming service).	No existing MEC attribute defined. See note.
svcContSupp	array(ACRScenario)	O	1..N	The ACR scenarios supported by the EAS for service continuity. If this attribute is not present, then the EAS does not support service continuity.	No existing MEC attribute defined. See note.
appLocs	array(RouteToLocation)	O	1..N	List of DNAI(s) and the corresponding N6 traffic routing information/routing profile ID, associated with the EAS. It is a subset of the DNAI(s) associated with the EDN where the EAS resides.	No existing MEC attribute defined. See note.

Attribute name	Data type	P	Cardinality	Description	Remarks
avlRep	DurationSec	O	0..1	The period indicating to the EES, how often the EES needs to check the EAS's availability after a successful registration.	No existing MEC attribute defined. See note.
status	string	O	0..1	EAS status (e.g. Enabled, Disabled etc.).	Similar to *state* attribute defined in the present document. See note.
NOTE: Such attributes can be made available to the MEC system via MEC application registration procedure. It is up to implementation and deployment on how such attributes are utilized in the MEC system.					

Annex D (informative): Change history

Date	Version	Information about changes
January 2021	V3.0.1	Updated with the agreements in MEC(20)00424r1.
November 2021	V3.0.2	Updated with the agreements in MEC(21)00430r6.
December 2021	V3.0.3	Updated with the agreements in MEC(21)00586r1.
December 2021	V3.0.4	Clean-up done by ***editHelp!*** E-mail: mailto:edithelp@etsi.org
January 2022	V3.0.5	Updated with the agreements in MEC(22)00037r2, MEC(22)00051r1 and MEC(22)00052.
March 2022	V3.0.6	Updated with the agreements in MEC(21)000557r8, MEC(22)000097r5, MEC(22)000113r1 and MEC(22)000114.
July 2022	V3.1.2	Updated with the agreements in MEC(22)000327. Rapporteur's clean-up.
July 2022	V3.1.3	Updated with the agreements in MEC(22)000348r1.
December 2022	V3.1.4	Updated with the agreements in MEC(22)000567r1, MEC(22)000568r1 and MEC(22)000590.
March 2023	V3.1.5	Updated with the agreements in MEC(23)000027r1.
March 2023	V3.1.6	Updated with the agreements in MEC(23)000078 and MEC(23)000102r1.
September 2023	V3.1.7	Updated with the agreements in MEC(23)000266r1.
September 2023	V3.1.8	Updated with the agreements in MEC(23)000262r2, MEC(23)000314r2, MEC(23)000315r2, MEC(23)000360, MEC(23)000367r1, MEC(23)000382r2 and MEC(23)000383r2.
December 2023	V3.1.9	Updated with the agreements in MEC(23)000522, MEC(23)000519r1, MEC(23)000521r2, MEC(23)000487r1, MEC(23)000542r1, MEC(23)000520r5, MEC(23)000523r2, MEC(23)000539, MEC(23)000540r1 and MEC(23)000541r1. Rapporteur's clean-up.
December 2023	V3.1.10	Clean-up done by ***editHelp!*** E-mail: edithelp@etsi.org
February 2024	V3.1.11	Rapporteur's clean-up.
March 2024	V3.1.12	Updated with the following text proposals to address the comments raised during the RC for Review: • MEC(24)000092 • MEC(24)000109r1 • MEC(24)000110 • MEC(24)000111 • MEC(24)000113r1 • MEC(24)000114r1 • MEC(24)000115r1 • MEC(24)000116 • MEC(24)000117 • MEC(24)000118 • MEC(24)000128r1 • MEC(24)000137r1 • fixed the bug in Table 8.1.5.2-1 • editorial change in 9.0: missing plural
June 2024	V3.2.2	Start of work
June 2024	V3.2.3	Updated with the agreements in MEC(24)000255r1 and MEC(24)000257r1.
September 2024	V3.2.4	Updated with the agreements in MEC(24)000328 and MEC(24)000359.
September 2024	V3.2.5	Final draft V3.2.5 is similar to Stable draft V3.2.4, and ready to go to MEC Remote Consensus for review.
October 2024	V3.2.6	Updated with the agreements in MEC(24)000392 to fix the bugs identified in the RC for review.

History

Document history		
V1.1.1	July 2017	Publication
V2.1.1	November 2019	Publication
V2.2.1	December 2020	Publication
V3.2.1	April 2024	Publication
V3.3.1	November 2024	Publication